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Dielectric resonator antenna modules

Abstract

An electronic device may be provided with an antenna module having a substrate. A phased antenna array of dielectric resonator antennas and a radio-frequency integrated circuit for the array may be mounted to one or more surfaces of the substrate. The dielectric resonator antennas may include dielectric columns excited by feed probes. The feed probes may be printed onto sidewalls of the dielectric columns or may be pressed against the sidewalls by biasing structures. A plastic substrate may be molded over each dielectric column and each of the feed probes in the array. The feed probes may cover multiple polarizations. The array may include elements for covering multiple frequency bands. The dielectric columns may be aligned a longitudinal axis and may be rotated at a non-zero and non-perpendicular angle with respect to the longitudinal axis.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 16/920,297, filed Jul. 2, 2020, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

(1) This relates generally to electronic devices and, more particularly, to electronic devices with wireless circuitry.

(2) Electronic devices often include wireless circuitry. For example, cellular telephones, computers, and other devices often contain antennas and wireless transceivers for supporting wireless communications.

(3) It may be desirable to support wireless communications in millimeter wave and centimeter wave communications bands. Millimeter wave communications, which are sometimes referred to as extremely high frequency (EHF) communications, and centimeter wave communications involve communications at frequencies of about 10-300 GHz. Operation at these frequencies may support high bandwidths but may raise significant challenges. For example, radio-frequency communications in millimeter and centimeter wave communications bands can be characterized by substantial attenuation and/or distortion during signal propagation through various mediums. The presence of conductive electronic device components can also make it difficult to incorporate circuitry for handling millimeter and centimeter wave communications into the electronic device. In addition, if care is not taken, manufacturing variations can undesirably limit the mechanical reliability and wireless performance of the antennas in the electronic device.

(4) It would therefore be desirable to be able to provide electronic devices with improved components for supporting millimeter and centimeter wave communications.

SUMMARY

(5) An electronic device may be provided with a housing, a display, and wireless circuitry. The housing may include peripheral conductive housing structures that run around a periphery of the device. The display may include a display cover layer mounted to the peripheral conductive housing structures. The wireless circuitry may include a phased antenna array that conveys radio-frequency signals in one or more frequency bands between 10 GHz and 300 GHz. The phased antenna array may convey the radio-frequency signals through the display cover layer or other dielectric cover layers in the device.

(6) The phased antenna array may include probe-fed dielectric resonator antennas. The phased antenna array and a radio-frequency integrated circuit (RFIC) for the phased antenna array may both be integrated into an antenna module. The antenna module may include an antenna module substrate. The RFIC may be surface-mounted to a first surface of the substrate whereas the probe-fed dielectric resonator antennas are mounted to a second surface of the substrate. Alternatively, the RFIC and probe-fed dielectric resonator antennas may be mounted to the same surface of the substrate. An over-mold structure may be provided over the RFIC. Additional phased antenna arrays may be mounted to the substrate if desired.

(7) Each of the probe-fed dielectric resonator antennas may include a dielectric resonating element mounted to a surface of the substrate. One or two feed probes may be coupled to sidewalls of the dielectric resonating element at the surface of the substrate to feed the dielectric resonating element. In one suitable arrangement, the feed probes may be formed from conductive traces that are patterned onto the sidewalls. In this arrangement, each dielectric resonating element may be formed on the antenna module at the same time, thereby minimizing mechanical variations to optimize mechanical and wireless performance of the module. The antenna module may be cut

from a substrate used to form multiple antenna modules for multiple devices to minimize manufacturing cost and complexity if desired.

(8) In another suitable arrangement, the feed probes may be formed from stamped sheet metal and may be pressed against the sidewalls by feed probe biasing structures that are molded over the feed probes and at least some of the dielectric resonating element. The feed probe biasing structures may also press parasitic elements against the sidewalls if desired. A plastic substrate may be molded over the feed probes and at least some of the dielectric resonating element for each of the antennas in the array to form an antenna package. The antenna package may be surface-mounted to the substrate (e.g., a flexible printed circuit) to form the antenna module. The antenna module may be aligned with a notch in a display module for the device. The dielectric resonating elements may be aligned along a longitudinal axis. If desired, each of the sidewalls of the dielectric resonating elements may be rotated at non-zero and non-perpendicular angles with respect to the longitudinal axis to maximize isolation between the antennas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an illustrative electronic device in accordance with some embodiments.

(2) FIG. 2 is a schematic diagram of illustrative circuitry in an electronic device in accordance with some embodiments.

(3) FIG. 3 is a schematic diagram of illustrative wireless circuitry in accordance with some embodiments.

(4) FIG. 4 is a diagram of an illustrative phased antenna array that may be adjusted using control circuitry to direct a beam of signals in accordance with some embodiments.

(5) FIG. 5 is a cross-sectional side view of an illustrative electronic device having phased antenna arrays for radiating through different sides of the device in accordance with some embodiments.

(6) FIG. 6 is a perspective view of an illustrative probe-fed dielectric resonator antenna for covering multiple polarizations in accordance with some embodiments.

(7) FIG. 7 is a top-down view of an illustrative probe-fed dielectric resonator antenna having multiple feed probes and floating parasitic patches for mitigating cross-polarization interference in accordance with some embodiments.

(8) FIG. 8 is a top-down view of an illustrative probe-fed dielectric resonating antenna having a single feed probe and grounded parasitic patches for mitigating cross-polarization interference in accordance with some embodiments.

(9) FIG. 9 is a top-down view of an illustrative antenna module having dielectric resonator antennas in accordance with some embodiments.

(10) FIG. 10 is a cross-sectional side view of an illustrative antenna module having dielectric resonator antennas in accordance with some embodiments.

(11) FIG. 11 is a perspective view of an illustrative antenna module having dielectric resonator antennas in accordance with some embodiments.

(12) FIG. 12 is a top-down view of an illustrative antenna module having dielectric resonator antennas and a radio-frequency integrated circuit mounted to the same side of a substrate in accordance with some embodiments.

(13) FIG. 13 is a side view of an illustrative antenna module having dielectric resonator antennas and a radio-frequency integrated circuit mounted to the same side of a substrate in accordance with some embodiments.

(14) FIG. 14 is a side view of an illustrative antenna module having dielectric resonator antennas on opposing sides of a substrate in accordance with some embodiments.

- (15) FIG. **15** is a cross-sectional side view of an illustrative antenna module having patch antennas and dielectric resonator antennas at opposing sides of a substrate in accordance with some embodiments.
- (16) FIGS. **16** and **17** are diagrams of an illustrative assembly process for an antenna module having dielectric resonator antennas mounted to a substrate in accordance with some embodiments.
- (17) FIG. **18** is a flow chart of illustrative steps that may be performed in assembling an antenna module having dielectric resonator antennas mounted to a substrate in accordance with some embodiments.
- (18) FIG. **19** is a perspective view of an illustrative antenna module having dielectric resonator antennas with feed probes that are biased towards dielectric resonating elements by biasing structures in accordance with some embodiments.
- (19) FIG. **20** is a diagram showing how an illustrative antenna module of the type shown in FIG. **19** may be assembled in accordance with some embodiments.
- (20) FIG. **21** is a top-down view of an illustrative electronic device having an antenna module aligned with a notch in a display module in accordance with some embodiments.
- (21) FIG. **22** is a top-down view of an illustrative antenna module having rotated dielectric resonating elements in accordance with some embodiments.
- (22) FIG. **23** is a perspective view of an illustrative antenna module having rotated dielectric resonating elements in accordance with some embodiments.
- (23) FIG. **24** is an exploded perspective view of an illustrative antenna module of the type shown in FIGS. **22** and **23** in accordance with some embodiments.

DETAILED DESCRIPTION

- (24) An electronic device such as electronic device **10** of FIG. **1** may contain wireless circuitry. The wireless circuitry may include one or more antennas. The antennas may include phased antenna arrays that are used for performing wireless communications using millimeter and centimeter wave signals. Millimeter wave signals, which are sometimes referred to as extremely high frequency (EHF) signals, propagate at frequencies above about 30 GHz (e.g., at 60 GHz or other frequencies between about 30 GHz and 300 GHz). Centimeter wave signals propagate at frequencies between about 10 GHz and 30 GHz. If desired, device **10** may also contain antennas for handling satellite navigation system signals, cellular telephone signals, local wireless area network signals, near-field communications, light-based wireless communications, or other wireless communications.
- (25) Electronic device **10** may be a portable electronic device or other suitable electronic device. For example, electronic device **10** may be a laptop computer, a tablet computer, a somewhat smaller device such as a wrist-watch device, pendant device, headphone device, earpiece device, or other wearable or miniature device, a handheld device such as a cellular telephone, a media player, or other small portable device. Device **10** may also be a set-top box, a desktop computer, a display into which a computer or other processing circuitry has been integrated, a display without an integrated computer, a wireless access point, a wireless base station, an electronic device incorporated into a kiosk, building, or vehicle, or other suitable electronic equipment.
- (26) Device **10** may include a housing such as housing **12**. Housing **12**, which may sometimes be referred to as a case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, etc.), other suitable materials, or a combination of these materials. In some situations, parts of housing **12** may be formed from dielectric or other low-conductivity material (e.g., glass, ceramic, plastic, sapphire, etc.). In other situations, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.
- (27) Device **10** may, if desired, have a display such as display **14**. Display **14** may be mounted on the front face of device **10** (e.g., display **14** may form some or all of the front face of the device). Display **14** may be a touch screen that incorporates capacitive touch electrodes or may be insensitive to touch. The rear face of housing **12** (i.e., the face of device **10** opposing the front face of device **10**) may have a substantially planar housing wall such as rear housing wall **12R** (e.g., a

planar housing wall). Rear housing wall **12R** may have slots that pass entirely through the rear housing wall and that therefore separate portions of housing **12** from each other. Rear housing wall **12R** may include conductive portions and/or dielectric portions. If desired, rear housing wall **12R** may include a planar metal layer covered by a thin layer or coating of dielectric such as glass, plastic, sapphire, or ceramic. Housing **12** may also have shallow grooves that do not pass entirely through housing **12**. The slots and grooves may be filled with plastic or other dielectrics. If desired, portions of housing **12** that have been separated from each other (e.g., by a through slot) may be joined by internal conductive structures (e.g., sheet metal or other metal members that bridge the slot).

(28) Housing **12** may include peripheral housing structures such as peripheral structures **12W**. Conductive portions of peripheral structures **12W** and conductive portions of rear housing wall **12R** may sometimes be referred to herein collectively as conductive structures of housing **12**. Peripheral structures **12W** may run around the periphery of device **10** and display **14**. In configurations in which device **10** and display **14** have a rectangular shape with four edges, peripheral structures **12W** may be implemented using peripheral housing structures that have a rectangular ring shape with four corresponding edges and that extend from rear housing wall **12R** to the front face of device **10** (as an example). Peripheral structures **12W** or part of peripheral structures **12W** may serve as a bezel for display **14** (e.g., a cosmetic trim that surrounds all four sides of display **14** and/or that helps hold display **14** to device **10**) if desired. Peripheral structures **12W** may, if desired, form sidewall structures for device **10** (e.g., by forming a metal band with vertical sidewalls, curved sidewalls, etc.).

(29) Peripheral structures **12W** may be formed of a conductive material such as metal and may therefore sometimes be referred to as peripheral conductive housing structures, conductive housing structures, peripheral metal structures, peripheral conductive sidewalls, peripheral conductive sidewall structures, conductive housing sidewalls, peripheral conductive housing sidewalls, sidewalls, sidewall structures, or a peripheral conductive housing member (as examples). Peripheral conductive housing structures **12W** may be formed from a metal such as stainless steel, aluminum, or other suitable materials. One, two, or more than two separate structures may be used in forming peripheral conductive housing structures **12W**.

(30) It is not necessary for peripheral conductive housing structures **12W** to have a uniform cross-section. For example, the top portion of peripheral conductive housing structures **12W** may, if desired, have an inwardly protruding ledge that helps hold display **14** in place. The bottom portion of peripheral conductive housing structures **12W** may also have an enlarged lip (e.g., in the plane of the rear surface of device **10**). Peripheral conductive housing structures **12W** may have substantially straight vertical sidewalls, may have sidewalls that are curved, or may have other suitable shapes. In some configurations (e.g., when peripheral conductive housing structures **12W** serve as a bezel for display **14**), peripheral conductive housing structures **12W** may run around the lip of housing **12** (i.e., peripheral conductive housing structures **12W** may cover only the edge of housing **12** that surrounds display **14** and not the rest of the sidewalls of housing **12**).

(31) Rear housing wall **12R** may lie in a plane that is parallel to display **14**. In configurations for device **10** in which some or all of rear housing wall **12R** is formed from metal, it may be desirable to form parts of peripheral conductive housing structures **12W** as integral portions of the housing structures forming rear housing wall **12R**. For example, rear housing wall **12R** of device **10** may include a planar metal structure and portions of peripheral conductive housing structures **12W** on the sides of housing **12** may be formed as flat or curved vertically extending integral metal portions of the planar metal structure (e.g., housing structures **12R** and **12W** may be formed from a continuous piece of metal in a unibody configuration). Housing structures such as these may, if desired, be machined from a block of metal and/or may include multiple metal pieces that are assembled together to form housing **12**. Rear housing wall **12R** may have one or more, two or more, or three or more portions. Peripheral conductive housing structures **12W** and/or conductive

portions of rear housing wall **12R** may form one or more exterior surfaces of device **10** (e.g., surfaces that are visible to a user of device **10**) and/or may be implemented using internal structures that do not form exterior surfaces of device **10** (e.g., conductive housing structures that are not visible to a user of device **10** such as conductive structures that are covered with layers such as thin cosmetic layers, protective coatings, and/or other coating layers that may include dielectric materials such as glass, ceramic, plastic, or other structures that form the exterior surfaces of device **10** and/or serve to hide peripheral conductive housing structures **12W** and/or conductive portions of rear housing wall **12R** from view of the user).

(32) Display **14** may have an array of pixels that form an active area **AA** that displays images for a user of device **10**. For example, active area **AA** may include an array of display pixels. The array of pixels may be formed from liquid crystal display (LCD) components, an array of electrophoretic pixels, an array of plasma display pixels, an array of organic light-emitting diode display pixels or other light-emitting diode pixels, an array of electrowetting display pixels, or display pixels based on other display technologies. If desired, active area **AA** may include touch sensors such as touch sensor capacitive electrodes, force sensors, or other sensors for gathering a user input.

(33) Display **14** may have an inactive border region that runs along one or more of the edges of active area **AA**. Inactive area **IA** of display **14** may be free of pixels for displaying images and may overlap circuitry and other internal device structures in housing **12**. To block these structures from view by a user of device **10**, the underside of the display cover layer or other layers in display **14** that overlap inactive area **IA** may be coated with an opaque masking layer in inactive area **IA**. The opaque masking layer may have any suitable color. Inactive area **IA** may include a recessed region such as notch **8** that extends into active area **AA**. Active area **AA** may, for example, be defined by the lateral area of a display module for display **14** (e.g., a display module that includes pixel circuitry, touch sensor circuitry, etc.). The display module may have a recess or notch in upper region **20** of device **10** that is free from active display circuitry (i.e., that forms notch **8** of inactive area **IA**). Notch **8** may be a substantially rectangular region that is surrounded (defined) on three sides by active area **AA** and on a fourth side by peripheral conductive housing structures **12W**.

(34) Display **14** may be protected using a display cover layer such as a layer of transparent glass, clear plastic, transparent ceramic, sapphire, or other transparent crystalline material, or other transparent layer(s). The display cover layer may have a planar shape, a convex curved profile, a shape with planar and curved portions, a layout that includes a planar main area surrounded on one or more edges with a portion that is bent out of the plane of the planar main area, or other suitable shapes. The display cover layer may cover the entire front face of device **10**. In another suitable arrangement, the display cover layer may cover substantially all of the front face of device **10** or only a portion of the front face of device **10**. Openings may be formed in the display cover layer. For example, an opening may be formed in the display cover layer to accommodate a button. An opening may also be formed in the display cover layer to accommodate ports such as speaker port **16** in notch **8** or a microphone port. Openings may be formed in housing **12** to form communications ports (e.g., an audio jack port, a digital data port, etc.) and/or audio ports for audio components such as a speaker and/or a microphone if desired.

(35) Display **14** may include conductive structures such as an array of capacitive electrodes for a touch sensor, conductive lines for addressing pixels, driver circuits, etc. Housing **12** may include internal conductive structures such as metal frame members and a planar conductive housing member (sometimes referred to as a backplate) that spans the walls of housing **12** (i.e., a substantially rectangular sheet formed from one or more metal parts that is welded or otherwise connected between opposing sides of peripheral conductive housing structures **12W**). The backplate may form an exterior rear surface of device **10** or may be covered by layers such as thin cosmetic layers, protective coatings, and/or other coatings that may include dielectric materials such as glass, ceramic, plastic, or other structures that form the exterior surfaces of device **10** and/or serve to hide the backplate from view of the user. Device **10** may also include conductive

structures such as printed circuit boards, components mounted on printed circuit boards, and other internal conductive structures. These conductive structures, which may be used in forming a ground plane in device **10**, may extend under active area AA of display **14**, for example.

(36) In regions **22** and **20**, openings may be formed within the conductive structures of device **10** (e.g., between peripheral conductive housing structures **12W** and opposing conductive ground structures such as conductive portions of rear housing wall **12R**, conductive traces on a printed circuit board, conductive electrical components in display **14**, etc.). These openings, which may sometimes be referred to as gaps, may be filled with air, plastic, and/or other dielectrics and may be used in forming slot antenna resonating elements for one or more antennas in device **10**, if desired.

(37) Conductive housing structures and other conductive structures in device **10** may serve as a ground plane for the antennas in device **10**. The openings in regions **22** and **20** may serve as slots in open or closed slot antennas, may serve as a central dielectric region that is surrounded by a conductive path of materials in a loop antenna, may serve as a space that separates an antenna resonating element such as a strip antenna resonating element or an inverted-F antenna resonating element from the ground plane, may contribute to the performance of a parasitic antenna resonating element, or may otherwise serve as part of antenna structures formed in regions **22** and **20**. If desired, the ground plane that is under active area AA of display **14** and/or other metal structures in device **10** may have portions that extend into parts of the ends of device **10** (e.g., the ground may extend towards the dielectric-filled openings in regions **22** and **20**), thereby narrowing the slots in regions **22** and **20**.

(38) In general, device **10** may include any suitable number of antennas (e.g., one or more, two or more, three or more, four or more, etc.). The antennas in device **10** may be located at opposing first and second ends of an elongated device housing (e.g., ends at regions **22** and **20** of device **10** of FIG. **1**), along one or more edges of a device housing, in the center of a device housing, in other suitable locations, or in one or more of these locations. The arrangement of FIG. **1** is merely illustrative.

(39) Portions of peripheral conductive housing structures **12W** may be provided with peripheral gap structures. For example, peripheral conductive housing structures **12W** may be provided with one or more gaps such as gaps **18**, as shown in FIG. **1**. The gaps in peripheral conductive housing structures **12W** may be filled with dielectric such as polymer, ceramic, glass, air, other dielectric materials, or combinations of these materials. Gaps **18** may divide peripheral conductive housing structures **12W** into one or more peripheral conductive segments. The conductive segments that are formed in this way may form parts of antennas in device **10** if desired. Gaps **18** may be omitted if desired. Other dielectric openings may be formed in peripheral conductive housing structures **12W** (e.g., dielectric openings other than gaps **18**) and may serve as dielectric antenna windows for antennas mounted within the interior of device **10**. Antennas within device **10** may be aligned with the dielectric antenna windows for conveying radio-frequency signals through peripheral conductive housing structures **12W**. Antennas within device **10** may also be aligned with inactive area IA of display **14** for conveying radio-frequency signals through display **14**.

(40) In order to provide an end user of device **10** with as large of a display as possible (e.g., to maximize an area of the device used for displaying media, running applications, etc.), it may be desirable to increase the amount of area at the front face of device **10** that is covered by active area AA of display **14**. Increasing the size of active area AA may reduce the size of inactive area IA within device **10**. This may reduce the area behind display **14** that is available for antennas within device **10**. For example, active area AA of display **14** may include conductive structures that serve to block radio-frequency signals handled by antennas mounted behind active area AA from radiating through the front face of device **10**. It would therefore be desirable to be able to provide antennas that occupy a small amount of space within device **10** (e.g., to allow for as large of a display active area AA as possible) while still allowing the antennas to communicate with wireless equipment external to device **10** with satisfactory efficiency bandwidth.

(41) In a typical scenario, device **10** may have one or more upper antennas and one or more lower antennas (as an example). An upper antenna may, for example, be formed at the upper end of device **10** in region **20**. A lower antenna may, for example, be formed at the lower end of device **10** in region **22**. Additional antennas may be formed along the edges of housing **12** extending between regions **20** and **22** if desired. The antennas may be used separately to cover identical communications bands, overlapping communications bands, or separate communications bands. The antennas may be used to implement an antenna diversity scheme or a multiple-input-multiple-output (MIMO) antenna scheme. Other antennas for covering any other desired frequencies may also be mounted at any desired locations within the interior of device **10**. The example of FIG. **1** is merely illustrative. If desired, housing **12** may have other shapes (e.g., a square shape, cylindrical shape, spherical shape, combinations of these and/or different shapes, etc.).

(42) A schematic diagram of illustrative components that may be used in device **10** is shown in FIG. **2**. As shown in FIG. **2**, device **10** may include control circuitry **28**. Control circuitry **28** may include storage such as storage circuitry **30**. Storage circuitry **30** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Control circuitry **28** may include processing circuitry such as processing circuitry **32**. Processing circuitry **32** may be used to control the operation of device **10**. Processing circuitry **32** may include on one or more microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, central processing units (CPUs), etc. Control circuitry **28** may be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on storage circuitry **30** (e.g., storage circuitry **30** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on storage circuitry **30** may be executed by processing circuitry **32**.

(43) Control circuitry **28** may be used to run software on device **10** such as internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **28** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **28** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols—sometimes referred to as WiFi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other WPAN protocols, IEEE 802.11ad protocols, cellular telephone protocols, MIMO protocols, antenna diversity protocols, satellite navigation system protocols, antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), etc. Each communication protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

(44) Device **10** may include input-output circuitry **24**. Input-output circuitry **24** may include input-output devices **26**. Input-output devices **26** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **26** may include user interface devices, data port devices, sensors, and other input-output components. For example, input-output devices may include touch screens, displays without touch sensor capabilities, buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, speakers, status indicators, light sources, audio jacks and other audio port components, digital data port devices, light sensors, gyroscopes, accelerometers or other components that can detect motion and device orientation relative to the Earth, capacitance sensors, proximity sensors

(e.g., a capacitive proximity sensor and/or an infrared proximity sensor), magnetic sensors, and other sensors and input-output components.

(45) Input-output circuitry **24** may include wireless circuitry such as wireless circuitry **34** for wirelessly conveying radio-frequency signals. While control circuitry **28** is shown separately from wireless circuitry **34** in the example of FIG. 2 for the sake of clarity, wireless circuitry **34** may include processing circuitry that forms a part of processing circuitry **32** and/or storage circuitry that forms a part of storage circuitry **30** of control circuitry **28** (e.g., portions of control circuitry **28** may be implemented on wireless circuitry **34**). As an example, control circuitry **28** may include baseband processor circuitry or other control components that form a part of wireless circuitry **34**.

(46) Wireless circuitry **34** may include millimeter and centimeter wave transceiver circuitry such as millimeter/centimeter wave transceiver circuitry **38**. Millimeter/centimeter wave transceiver circuitry **38** may support communications at frequencies between about 10 GHz and 300 GHz. For example, millimeter/centimeter wave transceiver circuitry **38** may support communications in Extremely High Frequency (EHF) or millimeter wave communications bands between about 30 GHz and 300 GHz and/or in centimeter wave communications bands between about 10 GHz and 30 GHz (sometimes referred to as Super High Frequency (SHF) bands). As examples, millimeter/centimeter wave transceiver circuitry **38** may support communications in an IEEE K communications band between about 18 GHz and 27 GHz, a K.sub.a communications band between about 26.5 GHz and 40 GHz, a K.sub.u communications band between about 12 GHz and 18 GHz, a V communications band between about 40 GHz and 75 GHz, a W communications band between about 75 GHz and 110 GHz, or any other desired frequency band between approximately 10 GHz and 300 GHz. If desired, millimeter/centimeter wave transceiver circuitry **38** may support IEEE 802.11ad communications at 60 GHz and/or 5th generation mobile networks or 5.sup.th generation wireless systems (5G) communications bands between 27 GHz and 90 GHz.

Millimeter/centimeter wave transceiver circuitry **38** may be formed from one or more integrated circuits (e.g., multiple integrated circuits mounted on a common printed circuit in a system-in-package device, one or more integrated circuits mounted on different substrates, etc.).

(47) If desired, millimeter/centimeter wave transceiver circuitry **38** (sometimes referred to herein simply as transceiver circuitry **38** or millimeter/centimeter wave circuitry **38**) may perform spatial ranging operations using radio-frequency signals at millimeter and/or centimeter wave signals that are transmitted and received by millimeter/centimeter wave transceiver circuitry **38**. The received signals may be a version of the transmitted signals that have been reflected off of external objects and back towards device **10**. Control circuitry **28** may process the transmitted and received signals to detect or estimate a range between device **10** and one or more external objects in the surroundings of device **10** (e.g., objects external to device **10** such as the body of a user or other persons, other devices, animals, furniture, walls, or other objects or obstacles in the vicinity of device **10**). If desired, control circuitry **28** may also process the transmitted and received signals to identify a two or three-dimensional spatial location of the external objects relative to device **10**.

(48) Spatial ranging operations performed by millimeter/centimeter wave transceiver circuitry **38** are unidirectional. Millimeter/centimeter wave transceiver circuitry **38** may additionally or alternatively perform bidirectional communications with external wireless equipment. Bidirectional communications involve both the transmission of wireless data by millimeter/centimeter wave transceiver circuitry **38** and the reception of wireless data that has been transmitted by external wireless equipment. The wireless data may, for example, include data that has been encoded into corresponding data packets such as wireless data associated with a telephone call, streaming media content, internet browsing, wireless data associated with software applications running on device **10**, email messages, etc.

(49) If desired, wireless circuitry **34** may include transceiver circuitry for handling communications at frequencies below 10 GHz such as non-millimeter/centimeter wave transceiver circuitry **36**. Non-millimeter/centimeter wave transceiver circuitry **36** may include wireless local area network

(WLAN) transceiver circuitry that handles 2.4 GHz and 5 GHz bands for Wi-Fi® (IEEE 802.11) communications, wireless personal area network (WPAN) transceiver circuitry that handles the 2.4 GHz Bluetooth® communications band, cellular telephone transceiver circuitry that handles cellular telephone communications bands from 700 to 960 MHz, 1710 to 2170 MHz, 2300 to 2700 MHz, and/or any other desired cellular telephone communications bands between 600 MHz and 4000 MHz, GPS receiver circuitry that receives GPS signals at 1575 MHz or signals for handling other satellite positioning data (e.g., GLONASS signals at 1609 MHz), television receiver circuitry, AM/FM radio receiver circuitry, paging system transceiver circuitry, ultra-wideband (UWB) transceiver circuitry, near field communications (NFC) circuitry, etc. Non-millimeter/centimeter wave transceiver circuitry **36** and millimeter/centimeter wave transceiver circuitry **38** may each include one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive radio-frequency components, switching circuitry, transmission line structures, and other circuitry for handling radio-frequency signals. Non-millimeter/centimeter wave transceiver circuitry **36** may be omitted if desired.

(50) Wireless circuitry **34** may include antennas **40**. Non-millimeter/centimeter wave transceiver circuitry **36** may convey radio-frequency signals below 10 GHz using one or more antennas **40**. Millimeter/centimeter wave transceiver circuitry **38** may convey radio-frequency signals above 10 GHz (e.g., at millimeter wave and/or centimeter wave frequencies) using antennas **40**. In general, transceiver circuitry **36** and **38** may be configured to cover (handle) any suitable communications (frequency) bands of interest. The transceiver circuitry may convey radio-frequency signals using antennas **40** (e.g., antennas **40** may convey the radio-frequency signals for the transceiver circuitry). The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external wireless communications equipment). Antennas **40** may transmit the radio-frequency signals by radiating the radio-frequency signals into free space (or to freespace through intervening device structures such as a dielectric cover layer). Antennas **40** may additionally or alternatively receive the radio-frequency signals from free space (e.g., through intervening devices structures such as a dielectric cover layer). The transmission and reception of radio-frequency signals by antennas **40** each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna by the radio-frequency signals within the frequency band(s) of operation of the antenna.

(51) In satellite navigation system links, cellular telephone links, and other long-range links, radio-frequency signals are typically used to convey data over thousands of feet or miles. In Wi-Fi® and Bluetooth® links at 2.4 and 5 GHz and other short-range wireless links, radio-frequency signals are typically used to convey data over tens or hundreds of feet. Millimeter/centimeter wave transceiver circuitry **38** may convey radio-frequency signals over short distances that travel over a line-of-sight path. To enhance signal reception for millimeter and centimeter wave communications, phased antenna arrays and beam steering techniques may be used (e.g., schemes in which antenna signal phase and/or magnitude for each antenna in an array are adjusted to perform beam steering). Antenna diversity schemes may also be used to ensure that the antennas that have become blocked or that are otherwise degraded due to the operating environment of device **10** can be switched out of use and higher-performing antennas used in their place.

(52) Antennas **40** in wireless circuitry **34** may be formed using any suitable antenna types. For example, antennas **40** may include antennas with resonating elements that are formed from stacked patch antenna structures, loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, monopole antenna structures, dipole antenna structures, helical antenna structures, Yagi (Yagi-Uda) antenna structures, hybrids of these designs, etc. In another suitable arrangement, antennas **40** may include antennas with dielectric resonating elements such as dielectric resonator antennas. If desired, one or more of antennas **40** may be cavity-backed antennas. Different types of antennas may be used for different

bands and combinations of bands. For example, one type of antenna may be used in forming a non-millimeter/centimeter wave wireless link for non-millimeter/centimeter wave transceiver circuitry **36** and another type of antenna may be used in conveying radio-frequency signals at millimeter and/or centimeter wave frequencies for millimeter/centimeter wave transceiver circuitry **38**. Antennas **40** that are used to convey radio-frequency signals at millimeter and centimeter wave frequencies may be arranged in one or more phased antenna arrays.

(53) A schematic diagram of an antenna **40** that may be formed in a phased antenna array for conveying radio-frequency signals at millimeter and centimeter wave frequencies is shown in FIG. **3**. As shown in FIG. **3**, antenna **40** may be coupled to millimeter/centimeter (MM/CM) wave transceiver circuitry **38**. Millimeter/centimeter wave transceiver circuitry **38** may be coupled to antenna feed **44** of antenna **40** using a transmission line path that includes radio-frequency transmission line **42**. Radio-frequency transmission line **42** may include a positive signal conductor such as signal conductor **46** and may include a ground conductor such as ground conductor **48**. Ground conductor **48** may be coupled to the antenna ground for antenna **40** (e.g., over a ground antenna feed terminal of antenna feed **44** located at the antenna ground). Signal conductor **46** may be coupled to the antenna resonating element for antenna **40**. For example, signal conductor **46** may be coupled to a positive antenna feed terminal of antenna feed **44** located at the antenna resonating element.

(54) In another suitable arrangement, antenna **40** may be a probe-fed antenna that is fed using a feed probe. In this arrangement, antenna feed **44** may be implemented as a feed probe. Signal conductor **46** may be coupled to the feed probe. Radio-frequency transmission line **42** may convey radio-frequency signals to and from the feed probe. When radio-frequency signals are being transmitted over the feed probe and the antenna, the feed probe may excite the resonating element for the antenna (e.g., may excite electromagnetic resonant modes of a dielectric antenna resonating element for antenna **40**). The resonating element may radiate the radio-frequency signals in response to excitation by the feed probe. Similarly, when radio-frequency signals are received by the antenna (e.g., from free space), the radio-frequency signals may excite the resonating element for the antenna (e.g., may excite electromagnetic resonant modes of the dielectric antenna resonating element for antenna **40**). This may produce antenna currents on the feed probe and the corresponding radio-frequency signals may be passed to the transceiver circuitry over the radio-frequency transmission line.

(55) Radio-frequency transmission line **42** may include a stripline transmission line (sometimes referred to herein simply as a stripline), a coaxial cable, a coaxial probe realized by metalized vias, a microstrip transmission line, an edge-coupled microstrip transmission line, an edge-coupled stripline transmission lines, a waveguide structure, combinations of these, etc. Multiple types of transmission lines may be used to form the transmission line path that couples millimeter/centimeter wave transceiver circuitry **38** to antenna feed **44**. Filter circuitry, switching circuitry, impedance matching circuitry, phase shifter circuitry, amplifier circuitry, and/or other circuitry may be interposed on radio-frequency transmission line **42**, if desired.

(56) Radio-frequency transmission lines in device **10** may be integrated into ceramic substrates, rigid printed circuit boards, and/or flexible printed circuits. In one suitable arrangement, radio-frequency transmission lines in device **10** may be integrated within multilayer laminated structures (e.g., layers of a conductive material such as copper and a dielectric material such as a resin that are laminated together without intervening adhesive) that may be folded or bent in multiple dimensions (e.g., two or three dimensions) and that maintain a bent or folded shape after bending (e.g., the multilayer laminated structures may be folded into a particular three-dimensional shape to route around other device components and may be rigid enough to hold its shape after folding without being held in place by stiffeners or other structures). All of the multiple layers of the laminated structures may be batch laminated together (e.g., in a single pressing process) without adhesive (e.g., as opposed to performing multiple pressing processes to laminate multiple layers together

with adhesive).

(57) FIG. 4 shows how antennas **40** for handling radio-frequency signals at millimeter and centimeter wave frequencies may be formed in a phased antenna array. As shown in FIG. 4, phased antenna array **54** (sometimes referred to herein as array **54**, antenna array **54**, or array **54** of antennas **40**) may be coupled to radio-frequency transmission lines **42**. For example, a first antenna **40-1** in phased antenna array **54** may be coupled to a first radio-frequency transmission line **42-1**, a second antenna **40-2** in phased antenna array **54** may be coupled to a second radio-frequency transmission line **42-2**, an Nth antenna **40-N** in phased antenna array **54** may be coupled to an Nth radio-frequency transmission line **42-N**, etc. While antennas **40** are described herein as forming a phased antenna array, the antennas **40** in phased antenna array **54** may sometimes also be referred to as collectively forming a single phased array antenna.

(58) Antennas **40** in phased antenna array **54** may be arranged in any desired number of rows and columns or in any other desired pattern (e.g., the antennas need not be arranged in a grid pattern having rows and columns). During signal transmission operations, radio-frequency transmission lines **42** may be used to supply signals (e.g., radio-frequency signals such as millimeter wave and/or centimeter wave signals) from millimeter/centimeter wave transceiver circuitry **38** (FIG. 3) to phased antenna array **54** for wireless transmission. During signal reception operations, radio-frequency transmission lines **42** may be used to supply signals received at phased antenna array **54** (e.g., from external wireless equipment or transmitted signals that have been reflected off of external objects) to millimeter/centimeter wave transceiver circuitry **38** (FIG. 3).

(59) The use of multiple antennas **40** in phased antenna array **54** allows beam steering arrangements to be implemented by controlling the relative phases and magnitudes (amplitudes) of the radio-frequency signals conveyed by the antennas. In the example of FIG. 4, antennas **40** each have a corresponding radio-frequency phase and magnitude controller **50** (e.g., a first phase and magnitude controller **50-1** interposed on radio-frequency transmission line **42-1** may control phase and magnitude for radio-frequency signals handled by antenna **40-1**, a second phase and magnitude controller **50-2** interposed on radio-frequency transmission line **42-2** may control phase and magnitude for radio-frequency signals handled by antenna **40-2**, an Nth phase and magnitude controller **50-N** interposed on radio-frequency transmission line **42-N** may control phase and magnitude for radio-frequency signals handled by antenna **40-N**, etc.).

(60) Phase and magnitude controllers **50** may each include circuitry for adjusting the phase of the radio-frequency signals on radio-frequency transmission lines **42** (e.g., phase shifter circuits) and/or circuitry for adjusting the magnitude of the radio-frequency signals on radio-frequency transmission lines **42** (e.g., power amplifier and/or low noise amplifier circuits). Phase and magnitude controllers **50** may sometimes be referred to collectively herein as beam steering circuitry (e.g., beam steering circuitry that steers the beam of radio-frequency signals transmitted and/or received by phased antenna array **54**).

(61) Phase and magnitude controllers **50** may adjust the relative phases and/or magnitudes of the transmitted signals that are provided to each of the antennas in phased antenna array **54** and may adjust the relative phases and/or magnitudes of the received signals that are received by phased antenna array **54**. Phase and magnitude controllers **50** may, if desired, include phase detection circuitry for detecting the phases of the received signals that are received by phased antenna array **54**. The term “beam” or “signal beam” may be used herein to collectively refer to wireless signals that are transmitted and received by phased antenna array **54** in a particular direction. The signal beam may exhibit a peak gain that is oriented in a particular pointing direction at a corresponding pointing angle (e.g., based on constructive and destructive interference from the combination of signals from each antenna in the phased antenna array). The term “transmit beam” may sometimes be used herein to refer to radio-frequency signals that are transmitted in a particular direction whereas the term “receive beam” may sometimes be used herein to refer to radio-frequency signals that are received from a particular direction. If, for example, phase and magnitude controllers **50**

are adjusted to produce a first set of phases and/or magnitudes for transmitted radio-frequency signals, the transmitted signals will form a transmit beam as shown by beam B1 of FIG. 4 that is oriented in the direction of point A. If, however, phase and magnitude controllers 50 are adjusted to produce a second set of phases and/or magnitudes for the transmitted signals, the transmitted signals will form a transmit beam as shown by beam B2 that is oriented in the direction of point B. Similarly, if phase and magnitude controllers 50 are adjusted to produce the first set of phases and/or magnitudes, radio-frequency signals (e.g., radio-frequency signals in a receive beam) may be received from the direction of point A, as shown by beam B1. If phase and magnitude controllers 50 are adjusted to produce the second set of phases and/or magnitudes, radio-frequency signals may be received from the direction of point B, as shown by beam B2.

(62) Each phase and magnitude controller 50 may be controlled to produce a desired phase and/or magnitude based on a corresponding control signal 52 received from control circuitry 28 of FIG. 2 (e.g., the phase and/or magnitude provided by phase and magnitude controller 50-1 may be controlled using control signal 52-1, the phase and/or magnitude provided by phase and magnitude controller 50-2 may be controlled using control signal 52-2, etc.). If desired, the control circuitry may actively adjust control signals 52 in real time to steer the transmit or receive beam in different desired directions over time. Phase and magnitude controllers 50 may provide information identifying the phase of received signals to control circuitry 28 if desired. A codebook on device 10 may map each beam pointing angle to a corresponding set of phase and magnitude values to be provided to phase and magnitude controllers 50 (e.g., the control circuitry may generate control signals 52 based on information from the codebook).

(63) When performing wireless communications using radio-frequency signals at millimeter and centimeter wave frequencies, the radio-frequency signals are conveyed over a line of sight path between phased antenna array 54 and external communications equipment. If the external object is located at point A of FIG. 4, phase and magnitude controllers 50 may be adjusted to steer the signal beam towards point A (e.g., to steer the pointing direction of the signal beam towards point A). Phased antenna array 54 may transmit and receive radio-frequency signals in the direction of point A. Similarly, if the external communications equipment is located at point B, phase and magnitude controllers 50 may be adjusted to steer the signal beam towards point B (e.g., to steer the pointing direction of the signal beam towards point B). Phased antenna array 54 may transmit and receive radio-frequency signals in the direction of point B. In the example of FIG. 4, beam steering is shown as being performed over a single degree of freedom for the sake of simplicity (e.g., towards the left and right on the page of FIG. 4). However, in practice, the beam may be steered over two or more degrees of freedom (e.g., in three dimensions, into and out of the page and to the left and right on the page of FIG. 4). Phased antenna array 54 may have a corresponding field of view over which beam steering can be performed (e.g., in a hemisphere or a segment of a hemisphere over the phased antenna array). If desired, device 10 may include multiple phased antenna arrays that each face a different direction to provide coverage from multiple sides of the device.

(64) FIG. 5 is a cross-sectional side view of device 10 in an example where device 10 has multiple phased antenna arrays. As shown in FIG. 5, peripheral conductive housing structures 12W may extend around the (lateral) periphery of device 10 and may extend from rear housing wall 12R to display 14. Display 14 may have a display module such as display module 64 (sometimes referred to as a display panel or conductive display structures). Display module 64 may include pixel circuitry, touch sensor circuitry, force sensor circuitry, and/or any other desired circuitry for forming active area AA of display 14. Display 14 may include a dielectric cover layer such as display cover layer 56 that overlaps display module 64. Display module 64 may emit image light and may receive sensor input through display cover layer 56. Display cover layer 56 and display 14 may be mounted to peripheral conductive housing structures 12W. The lateral area of display 14 that does not overlap display module 64 may form inactive area IA of display 14.

(65) Device 10 may include multiple phased antenna arrays (e.g., phased antenna arrays 54 of FIG.

4). For example, device **10** may include a rear-facing phased antenna array. The rear-facing phased antenna array may be adhered to rear housing wall **12R** using adhesive, may be pressed against (e.g., in contact with) rear housing wall **12R**, or may be spaced apart from rear housing wall **12R**. The rear-facing phased antenna array may transmit and/or receive radio-frequency signals **60** at millimeter and centimeter wave frequencies through rear housing wall **12R**. In scenarios where rear housing wall **12R** includes metal portions, radio-frequency signals **60** may be conveyed through an aperture or opening in the metal portions of rear housing wall **12R** or may be conveyed through other dielectric portions of rear housing wall **12R**. The aperture may be overlapped by a dielectric cover layer or dielectric coating that extends across the lateral area of rear housing wall **12R** (e.g., between peripheral conductive housing structures **12W**). The rear-facing phased antenna array may perform beam steering for radio-frequency signals **60** across at least some of the hemisphere below the rear face of device **10**.

(66) The field of view of the rear-facing phased antenna array is limited to the hemisphere under the rear face of device **10**. Display module **64** and other components **58** (e.g., portions of input-output circuitry **24** or control circuitry **28** of FIG. 2, a battery for device **10**, etc.) in device **10** include conductive structures. If care is not taken, these conductive structures may block radio-frequency signals from being conveyed by a phased antenna array within device **10** across the hemisphere over the front face of device **10**. While a front-facing phased antenna array for covering the hemisphere over the front face of device **10** may be mounted against display cover layer **56** within inactive area **IA**, there may be insufficient space between the lateral periphery of display module **64** and peripheral conductive housing structures **12W** to form all of the circuitry and radio-frequency transmission lines necessary to fully support the phased antenna array, particularly as the size of active area **AA** is maximized.

(67) In order to mitigate these issues and provide coverage through the front face of device **10**, a front-facing phased antenna array may be mounted within peripheral region **66** of device **10**. The antennas in the front-facing phased antenna array may include dielectric resonator antennas. Dielectric resonator antennas may occupy less area in the X-Y plane of FIG. 5 than other types of antennas such as patch antennas and slot antennas. Implementing the antennas as dielectric resonator antennas may allow the radiating elements of the front-facing phased antenna array to fit within inactive area **IA** between display module **64** and peripheral conductive housing structures **12W**. At the same time, the radio-frequency transmission lines and other components for the phased antenna array may be located behind (under) display module **64**. The front-facing phased antenna array may transmit and/or receive radio-frequency signals **62** at millimeter and centimeter wave frequencies through display cover layer **56**. The front-facing phased antenna array may perform beam steering for radio-frequency signals **62** across at least some of the hemisphere above the front face of device **10**.

(68) Device **10** may include both a front-facing phased antenna array (e.g., within peripheral region **66**) and a rear-facing phased antenna array (e.g., within peripheral region **66** or elsewhere between display module **64** and rear housing wall **12R**). If desired, device **10** may additionally or alternatively include one or more side-facing phased antenna arrays. The side-facing phased antenna arrays may be aligned with dielectric antenna windows in peripheral conductive housing structures **12W**. The front, rear, and/or side-facing phased antenna arrays may be omitted if desired. The front and rear-facing phased antenna arrays (and optionally the side-facing phased antenna arrays) may collectively provide radio-frequency cover across an entire sphere around device **10**.

(69) The phased antenna array(s) **54** in device **10** may be formed in corresponding integrated antenna modules. Each antenna module may include a substrate such as a rigid printed circuit board substrate, a flexible printed circuit substrate, a plastic substrate, or a ceramic substrate, and one or more phased antenna arrays mounted to the substrate. Each antenna module may also include electronic components (e.g., radio-frequency components) that support the operations of the phased antenna array(s) therein. For example, each antenna module may include a radio-frequency

integrated circuit (e.g., an integrated circuit chip) or other circuitry mounted to the corresponding substrate. Transmission line structures (e.g., radio-frequency signal traces), conductive vias, conductive traces, solder balls, or other conductive interconnect structures may couple the radio-frequency integrated circuit to each of the antennas in the phased antenna array(s) of the antenna module. The radio-frequency integrated circuit (RFIC) and/or other electronic components in the antenna module may include radio-frequency components such as amplifier circuitry, phase shifter circuitry (e.g., phase and magnitude controllers **50** of FIG. **4**), and/or other circuitry that operates on radio-frequency signals. The rear-facing, front-facing, and/or side-facing phased antenna array(s) in device **10** may be formed within respective antenna modules. In another suitable arrangement, a rear-facing and front-facing phased antenna array may be formed as a part of the same antenna module in device **10**.

(70) FIG. **6** is a perspective view of an illustrative probe-fed dielectric resonator antenna that may be used in forming the antennas of any of the phased antenna arrays in device **10**. Antenna **40** of FIG. **6** may be a dielectric resonator antenna. In this example, antenna **40** includes a dielectric resonating element **68** mounted to an underlying substrate such as substrate **72**. Substrate **72** may, for example, be the substrate of a corresponding antenna module in device **10**. Substrate **72** may be a rigid printed circuit board substrate, a flexible printed circuit substrate, a ceramic substrate, a plastic substrate, or any other desired substrate.

(71) In the example of FIG. **6**, antenna **40** is a dual-polarization antenna that conveys both vertically and horizontally polarized radio-frequency signals **84** (e.g., linearly-polarized signals having orthogonal electric field orientations). This example is merely illustrative and, in another suitable arrangement, antenna **40** may only cover a single polarization. Antenna **40** may be fed using radio-frequency transmission lines that are formed on and/or embedded within flexible substrate **72** such as radio-frequency transmission lines **88** (e.g., a first radio-frequency transmission line **88V** for conveying vertically-polarized signals and a second radio-frequency transmission line **88H** for conveying horizontally-polarized signals). Radio-frequency transmission lines **88V** and **88H** may, for example, form part of radio-frequency transmission lines **42** of FIGS. **3** and **4**. Radio-frequency transmission lines **88V** and **88H** may include ground traces (e.g., for forming part of ground conductor **48** of FIG. **3**) and signal traces (e.g., for forming part of signal conductor **46** of FIG. **3**) on and/or embedded within substrate **72**. Radio-frequency transmission lines **88V** and **88H** may be coupled to a radio-frequency integrated circuit or other radio-frequency components on the antenna module that includes antenna **40**.

(72) Dielectric resonating element **68** of antenna **40** may be formed from a column (pillar) of dielectric material mounted to the top surface of substrate **72**. If desired, dielectric resonating element **68** may be embedded within (e.g., laterally surrounded by) a dielectric substrate mounted to the top surface of substrate **72** such as dielectric substrate **70**. Dielectric resonating element **68** may have a height **96** that extends from a bottom surface **82** at substrate **72** to an opposing top surface **80**. Dielectric substrate **70** (sometimes referred to herein as over-mold structure **70**) may extend across some or all of height **96**. Top surface **80** may lie flush with the top surface of dielectric substrate **70**, may protrude beyond the top surface of dielectric substrate **70**, or dielectric substrate **70** may extend over and cover top surface **80** of dielectric resonating element **68**.

(73) The operating (resonant) frequency of antenna **40** may be selected by adjusting the dimensions of dielectric resonating element **68** (e.g., in the direction of the X, Y, and/or Z axes of FIG. **6**). Dielectric resonating element **68** may be formed from a column of dielectric material having dielectric constant dk_1 . Dielectric constant dk_1 may be relatively high (e.g., greater than 10.0, greater than 12.0, greater than 15.0, greater than 20.0, between 22.0 and 25.0, between 15.0 and 40.0, between 10.0 and 50.0, between 18.0 and 30.0, between 12.0 and 45.0, etc.). In one suitable arrangement, dielectric resonating element **68** may be formed from zirconia or a ceramic material. Other dielectric materials may be used to form dielectric resonating element **68** if desired.

(74) Dielectric substrate **70** may be formed from a material having dielectric constant dk_2 .

Dielectric constant dk_2 may be less than dielectric constant dk_1 of dielectric resonating element **68** (e.g., less than 18.0, less than 15.0, less than 10.0, between 3.0 and 4.0, less than 5.0, between 2.0 and 5.0, etc.). Dielectric constant dk_2 may be less than dielectric constant dk_1 by at least 10.0, 5.0, 15.0, 12.0, 6.0, etc. In one suitable arrangement, dielectric substrate **70** may be formed from molded plastic (e.g., injection molded plastic). Other dielectric materials may be used to form dielectric substrate **70** or dielectric substrate **70** may be omitted if desired. The difference in dielectric constant between dielectric resonating element **68** and dielectric substrate **70** may establish a radio-frequency boundary condition between dielectric resonating element **68** and dielectric substrate **70** from bottom surface **82** to top surface **80**. This may configure dielectric resonating element **68** to serve as a resonating waveguide for propagating radio-frequency signals **84** at millimeter and centimeter wave frequencies.

(75) Dielectric substrate **70** may have a width (thickness) **94** on some or all sides of dielectric resonating element **68**. Width **94** may be selected to isolate dielectric resonating element **68** from surrounding device structures and/or from other dielectric resonating elements in the same antenna module and to minimize signal reflections in dielectric substrate **70**. Width **94** may be, for example, at least one-tenth of the effective wavelength of the radio-frequency signals in a dielectric material of dielectric constant dk_2 . Width **94** may be 0.4-0.5 mm, 0.3-0.5 mm, 0.2-0.6 mm, greater than 0.1 mm, greater than 0.3 mm, 0.2-2.0 mm, 0.3-1.0 mm, or greater than between 0.4 and 0.5 mm, just as a few examples.

(76) Dielectric resonating element **68** may radiate radio-frequency signals **84** when excited by the signal conductor for radio-frequency transmission lines **88V** and/or **88H**. In some scenarios, a slot is formed in ground traces on substrate **72**, the slot is indirectly fed by a signal conductor embedded within substrate **72**, and the slot excites dielectric resonating element **68** to radiate radio-frequency signals **84**. However, in these scenarios, the radiating characteristics of the antenna may be affected by how the dielectric resonating element is mounted to substrate **72**. For example, air gaps or layers of adhesive used to mount the dielectric resonating element to the flexible printed circuit can be difficult to control and can undesirably affect the radiating characteristics of the antenna. In order to mitigate the issues associated with exciting dielectric resonating element **68** using an underlying slot, antenna **40** may be fed using one or more radio-frequency feed probes **100** such as feed probes **100V** and **100H** of FIG. 6. Feed probes **100** may form part of the antenna feeds for antenna **40** (e.g., antenna feed **44** of FIG. 3).

(77) As shown in FIG. 6, feed probe **100V** may be formed from conductive structure **86V** and feed probe **100H** may be formed from conductive structure **86H**. Conductive structure **86V** may include a first portion patterned onto or pressed against a first sidewall **102** of dielectric resonating element **68**. If desired, conductive structure **86V** may also include a second portion on the surface of substrate **72** and the second portion may be coupled to the signal traces of radio-frequency transmission line **88V** (e.g., using solder, welds, conductive adhesive, etc.). The second portion of conductive structure **86V** may be omitted if desired (e.g., the signal traces in radio-frequency transmission line **88V** may be soldered directly to the portion of conductive structure **86V** on the first sidewall **102**). Conductive structure **86V** may include conductive traces patterned directly onto the first sidewall **102** or may include stamped sheet metal in scenarios where conductive structure **86V** is pressed against the first sidewall **102**, as examples.

(78) The signal traces in radio-frequency transmission line **88V** may convey radio-frequency signals to and from feed probe **100V**. Feed probe **100V** may electromagnetically couple the radio-frequency signals on the signal traces of radio-frequency transmission line **88V** into dielectric resonating element **68**. This may serve to excite one or more electromagnetic modes (e.g., radio-frequency cavity or waveguide modes) of dielectric resonating element **68**. When excited by feed probe **100V**, the electromagnetic modes of dielectric resonating element **68** may configure the dielectric resonating element to serve as a waveguide that propagates the wavefronts of radio-frequency signals **84** along the height of dielectric resonating element **68** (e.g., in the direction of

the Z-axis and along the central/longitudinal axis **76** of dielectric resonating element **68**). The radio-frequency signals **84** conveyed by feed probe **100V** may be vertically polarized.

(79) Similarly, conductive structure **86H** may include a first portion patterned onto or pressed against a second sidewall **102** of dielectric resonating element **68**. If desired, conductive structure **86H** may also include a second portion on the surface of substrate **72** and the second portion may be coupled to the signal traces of radio-frequency transmission line **88H** (e.g., using solder, welds, conductive adhesive, etc.). The second portion of conductive structure **86H** may be omitted if desired (e.g., the signal traces in radio-frequency transmission line **88H** may be soldered directly to the conductive structure **86H** on sidewall **102**). Conductive structure **86H** may include conductive traces patterned directly onto the second sidewall **102** or may include stamped sheet metal in scenarios where conductive structure **86H** is pressed against the second sidewall **102**, as examples.

(80) The signal traces in radio-frequency transmission line **88H** may convey radio-frequency signals to and from feed probe **100H**. Feed probe **100H** may electromagnetically couple the radio-frequency signals on the signal traces of radio-frequency transmission line **88H** into dielectric resonating element **68**. This may serve to excite one or more electromagnetic modes (e.g., radio-frequency cavity or waveguide modes) of dielectric resonating element **68**. When excited by feed probe **100H**, the electromagnetic modes of dielectric resonating element **68** may configure the dielectric resonating element to serve as a waveguide that propagates the wavefronts of radio-frequency signals **84** along the height of dielectric resonating element **68** (e.g., along central/longitudinal axis **76** of dielectric resonating element **68**). The radio-frequency signals **84** conveyed by feed probe **100H** may be horizontally polarized.

(81) Similarly, during signal reception, radio-frequency signals **84** may be received by antenna **40**. The received radio-frequency signals may excite the electromagnetic modes of dielectric resonating element **68**, resulting in the propagation of the radio-frequency signals down the height of dielectric resonating element **68**. Feed probe **100V** may couple the received vertically-polarized signals onto radio-frequency transmission line **88V**. Feed probe **100H** may couple the received horizontally-polarized signals onto radio-frequency transmission line **88H**. Radio-frequency transmission lines **88H** and **88V** may pass the received radio-frequency signals to millimeter/centimeter wave transceiver circuitry (e.g., millimeter/centimeter wave transceiver circuitry **38** of FIGS. 2 and 3) through the radio-frequency integrated circuit for antenna **40**. The relatively large difference in dielectric constant between dielectric resonating element **68** and dielectric substrate **70** may allow dielectric resonating element **68** to convey radio-frequency signals **84** with a relatively high antenna efficiency (e.g., by establishing a strong boundary between dielectric resonating element **68** and dielectric substrate **70** for the radio-frequency signals). The relatively high dielectric constant of dielectric resonating element **68** may also allow the dielectric resonating element **68** to occupy a relatively small volume compared to scenarios where materials with a lower dielectric constant are used.

(82) The dimensions of feed probes **100V** and **100H** (e.g., height **90** and width **92** on sidewalls **102**) may be selected to help match the impedance of radio-frequency transmission lines **88V** and **88H** to the impedance of dielectric resonating element **68**. As an example, width **92** may be between 0.3 mm and 0.7 mm, between 0.2 mm and 0.8 mm, between 0.4 mm and 0.6 mm, or other values. Height **90** may be between 0.3 mm and 0.7 mm, between 0.2 mm and 0.8 mm, between 0.4 mm and 0.6 mm, or other values. Height **90** may be equal to width **92** or may be different than width **92**. Feed probes **100V** and **100H** may sometimes be referred to herein as feed conductors, feed patches, or probe feeds. Dielectric resonating element **68** may sometimes be referred to herein as a dielectric radiating element, dielectric radiator, dielectric resonator, dielectric antenna resonating element, dielectric column, dielectric pillar, radiating element, or resonating element. When fed by one or more feed probes such as feed probes **100V** and **100H**, dielectric resonator antennas such as antenna **40** of FIG. 6 may sometimes be referred to herein as probe-fed dielectric resonator antennas.

(83) Antenna **40** may be included in a rear-facing, front-facing, or side-facing phased antenna array in device **10** (e.g., radio-frequency signals **84** may form radio-frequency signals **62** or **60** of FIG. 5). In scenarios where antenna **40** is formed in a front-facing phased antenna array, top surface **80** may be pressed against, adhered to, or separated from display cover layer **56** of FIG. 5. In scenarios where antenna **40** is formed in a rear-facing phased antenna array, top surface **80** may be pressed against, adhered to, or separated from rear housing wall **12R** of FIG. 5. An optional impedance matching layer may be interposed between top surface **80** and rear housing wall **12R** or display cover layer **56**. The impedance matching layer may have a dielectric constant that is between dielectric constant dk_1 and the dielectric constant of rear housing wall **12R** or display cover layer **56**. If desired, the dielectric constant and thickness of the impedance matching layer may be selected to configure the impedance matching layer to form a quarter-wave impedance transformer for antenna **40** at the frequencies of operation of antenna **40**. This may configure the impedance matching layer to help minimize signal reflections at the interfaces between top surface **80** and free space exterior to device **10**.

(84) If desired, radio-frequency transmission lines **88V** and **88H** may include impedance matching structures (e.g., transmission line stubs) to help match the impedance of dielectric resonating element **68**. Both feed probes **100H** and **100V** may be active at once so that antenna **40** conveys both vertically and horizontally polarized signals at any given time. If desired, the phases of the signals conveyed by feed probes **100H** and **100V** may be independently adjusted so that antenna **40** conveys radio-frequency signals **84** with an elliptical or circular polarization. In another suitable arrangement, a single one of feed probes **100H** and **100V** may be active at once so that antenna **40** conveys radio-frequency signals of only a single polarization at any given time. In another suitable arrangement, antenna **40** may be a single-polarization antenna where radio-frequency transmission line **88V** and feed probe **100V** have been omitted.

(85) As shown in FIG. 6, dielectric resonating element **68** may have a height **96**, a length **74**, and a width **73**. Length **74**, width **73**, and height **96** may be selected to provide dielectric resonating element **68** with a corresponding mix of electromagnetic cavity/waveguide modes that, when excited by feed probes **100H** and/or **100V**, configure antenna **40** to radiate at desired frequencies. For example, height **96** may be 2-10 mm, 4-6 mm, 3-7 mm, 4.5-5.5 mm, or greater than 2 mm. Width **73** and length **74** may each be 0.5-1.0 mm, 0.4-1.2 mm, 0.7-0.9 mm, 0.5-2.0 mm, 1.5 mm-2.5 mm, 1.7 mm-1.9 mm, 1.0 mm-3.0 mm, etc. Width **73** may be equal to length **74** (e.g., dielectric resonating element **68** may have a square-shaped lateral profile in the X-Y plane) or, in other arrangements, may be different than length **74** (e.g., dielectric resonating element **68** may have a rectangular or non-rectangular lateral profile in the X-Y plane). Sidewalls **102** of dielectric resonating element **68** may directly contact the surrounding dielectric substrate **70**. Dielectric substrate **70** may be molded over feed probes **100H** and **100V** or may include openings, notches, or other structures that accommodate the presence of feed probes **100H** and **100V**. Each sidewall **102** may be planar or, if desired, one or more sidewall **102** may have a non-planar shape (e.g., a shape with planar and curved portions, a planar shape with a notch or recessed portion, etc.). The example of FIG. 6 is merely illustrative and, if desired, dielectric resonating element **68** may have other shapes (e.g., shapes with any desired number of straight and/or curved sidewalls **102**).

(86) In practice, if care is not taken, dielectric resonator antennas such as antenna **40** can be subject to undesirable cross-polarization interference. Cross-polarization interference can occur when radio-frequency signals to be conveyed in a first polarization are undesirably transmitted or received using an antenna feed that is used to convey radio-frequency signals in a second polarization. For example, cross-polarization interference may involve the leakage of horizontally-polarized signals onto feed probe **100V** of FIG. 6 (e.g., a feed probe intended to convey vertically-polarized signals) and/or the leakage of vertically-polarized signals onto feed probe **100H** of FIG. 6 (e.g., a feed probe intended to convey horizontally-polarized signals). The cross-polarization interference can arise when the electric field produced by feed probe **100V** has components

oriented at a mix of different angles or when the electric field produced by feed probe **100H** has components oriented at a mix of different angles within dielectric resonating element **68**. Cross-polarization interference can lead to a decrease in overall data throughput, errors in the transmitted or received data, or otherwise degraded antenna performance. These effects are also particularly detrimental in scenarios where antenna **40** conveys independent data streams using horizontal and vertical polarizations (e.g., under a MIMO scheme), as the cross-polarization interference reduces the independence of the data streams. It would therefore be desirable to be able to provide a dielectric resonator antenna such as antenna **40** with structures for mitigating cross polarization interference (e.g., for maximizing isolation between polarizations handled by the antenna).

(87) FIG. 7 is a top-down view of antenna **40** having structures for mitigating cross polarization interference. In the example of FIG. 7, antenna **40** is a dual-polarization dielectric resonator antenna having feed probes **100V** and **100H** for exciting different polarizations of dielectric resonating element **68**.

(88) As shown in FIG. 7, dielectric resonating element **68** may have a rectangular lateral profile. Dielectric resonating element **68** may have four sidewalls **102** (e.g., four vertical faces or surfaces oriented perpendicular to the X-Y plane) such as a first sidewall **102A**, a second sidewall **102B**, a third sidewall **102C**, and a fourth sidewall **102D**. Third sidewall **102C** may oppose first sidewall **102A** and fourth sidewall **102D** may oppose second sidewall **102B** on dielectric resonating element **68**. Conductive structure **86V** of feed probe **100V** may be patterned onto or pressed against first sidewall **102A**. Conductive structure **86V** may also be coupled to conductive trace **106V** on the underlying substrate **72** (e.g., using solder, welds, conductive adhesive, etc.). Conductive trace **106V** may be coupled to conductive trace **104V**. Conductive traces **104V** and **106V** may form part of the signal conductor for radio-frequency transmission line **88V** of FIG. 6. Similarly, conductive structure **86H** of feed probe **100H** may be patterned onto or pressed against second sidewall **102B**. Conductive structure **86H** may also be coupled to conductive trace **106H** on substrate **72** (e.g., using solder, welds, conductive adhesive, etc.). Conductive traces **106H** may be coupled to conductive trace **104H**. Conductive traces **104H** and **106H** may form part of the signal conductor for radio-frequency transmission line **88H** of FIG. 6.

(89) In order to mitigate cross polarization interference, parasitic elements such as parasitic elements **108H** and **108V** may be patterned onto the sidewalls of dielectric resonating element **68**. Parasitic elements **108H** and **108V** may, for example, be formed from floating patches of conductive material patterned onto or pressed against the sidewalls of dielectric resonating element **68** (e.g., conductive patches that are not coupled to ground or the signal traces for antenna **40**). As shown in FIG. 7, parasitic element **108H** may be patterned onto or pressed against fourth sidewall **102D** opposite feed probe **100H**. Parasitic element **108V** may be patterned onto or pressed against third sidewall **102C** opposite first feed probe **100V**.

(90) The presence of the conductive material in parasitic element **108H** may serve to change the boundary condition for the electric field excited by feed probe **100H** within dielectric resonating element **68**. For example, in scenarios where parasitic element **108H** is omitted, the electric field excited by feed probe **100H** may include a mix of different electric field components oriented in different directions. This may lead to cross-polarization interference in which some vertically-polarized signals undesirably leak onto feed probe **100H**. However, the boundary condition created by parasitic element **108H** may serve to align the electric field excited by feed probe **100H** in a single direction between sidewalls **102B** and **102D**, as shown by arrows **112** (e.g., in a horizontal direction parallel to the X-axis). Because the entire electric field excited by feed probe **100H** is horizontal, feed probe **100H** may only convey horizontally-polarized signals without vertically-polarized signals interfering with the horizontally-polarized signals.

(91) Similarly, the presence of the conductive material in parasitic element **108V** may serve to change the boundary condition for the electric field excited by feed probe **100V** within dielectric resonating element **68**. For example, in scenarios where parasitic element **108V** is omitted, the

electric field excited by feed probe **100V** may include a mix of different electric field components oriented in different directions. This may lead to cross-polarization interference in which some horizontally-polarized signals undesirably leak onto feed probe **100V**. However, the boundary condition created by parasitic element **108V** may serve to align the electric field excited by feed probe **100V** in a single direction between sidewalls **102A** and **102C**, as shown by arrows **110** (e.g., in a vertical direction parallel to the Y-axis). Because the entire electric field excited by feed probe **100V** is vertical, feed probe **100V** may only convey vertically-polarized signals without horizontally-polarized signals interfering with the vertically-polarized signals.

(92) Parasitic element **108V** may have a shape (e.g., lateral dimensions in the X-Z plane) that matches the shape of the portion of conductive structure **86V** on sidewall **102A** (e.g., parasitic element **108V** may have width **92** and height **90** of FIG. **6**). Similarly, parasitic element **108H** may have a shape (e.g., lateral dimensions in the Y-Z plane) that matches the shape of the portion of conductive structure **86H** on sidewall **102B** (e.g., parasitic element **108H** may have width **92** and height **90** of FIG. **6**). This may ensure that there are symmetric boundary conditions between feed probe **100V** and parasitic element **108V** and between feed probe **100H** and parasitic element **108H**. Parasitic element **108V** need not have the same exact dimensions as feed probe **100V** and parasitic element **108H** need not have the same exact dimensions as feed probe **100H** if desired.

(93) Antenna **40** may also include cross-polarization interference mitigating parasitic elements in scenarios where antenna **40** is fed using only a single feed probe. FIG. **8** is a top-down view showing how antenna **40** may include cross-polarization interference mitigating parasitic elements in an arrangement where antenna **40** is fed using only a single feed probe **100**.

(94) As shown in FIG. **8**, antenna **40** may be fed using a single feed probe **100**. Conductive structure **86** of feed probe **100** may be patterned onto sidewall **102A** of dielectric resonating element **68**. Conductive structure **86** may be coupled to conductive trace **104** on the underlying substrate **72**. Ground traces such as ground traces **116** may also be patterned onto substrate **72**.

(95) Antenna **40** may include one or more parasitic elements **114** such as a first parasitic element **114-1** and a second parasitic element **114-2**. Parasitic element **114-1** may be formed from a patch of conductive traces (e.g., a conductive patch) that is patterned onto sidewall **102D** of dielectric resonating element **68**. Parasitic element **114-2** may be formed from a patch of conductive traces (e.g., a conductive patch) that is patterned onto sidewall **102B** of dielectric resonating element **68**. Parasitic elements **114-1** and **114-2** may each have the same size and lateral dimensions (e.g., in the Y-Z plane) as conductive structure **86** (e.g., in the X-Z plane), for example. Parasitic element **114-1** and parasitic element **114-2** may each be coupled to ground traces **116** at substrate **72** by conductive interconnect structures **118**. Conductive interconnect structures **118** may include solder, welds, conductive adhesive, conductive tape, conductive foam, conductive springs, conductive brackets, and/or any other desired conductive interconnect structures. In this way, parasitic elements **114-1** and **114-2** may each be held at a ground potential (e.g., parasitic elements **114-1** and **114-2** may be grounded patches). Parasitic element **114-1** may be omitted or parasitic element **114-2** may be omitted if desired (e.g., antenna **40** may include only a single parasitic element **114** if desired).

(96) Parasitic element **114-1** and/or parasitic element **114-2** may serve to alter the electromagnetic boundary conditions of dielectric resonating element **68** to mitigate cross-polarization interference for feed probe **100** (e.g., to isolate feed probe **100** from interference from horizontally-polarized signals in scenarios where feed probe **100** handles vertically-polarized signals). Sidewall **102C** of dielectric resonating element **68** may be free from conductive material such as parasitic elements **114**.

(97) Phased antenna array **54** of FIG. **4** (e.g., a front-facing phased antenna array for conveying radio-frequency signals **62** through display cover layer **56** of FIG. **5**, a rear-facing phased antenna array for conveying radio-frequency signals **60** through rear housing wall **12R** of FIG. **5**, or a side-facing phased antenna array) may include any desired number of antennas **40** arranged in any

desired pattern (e.g., a pattern having rows and columns). Each of the antennas **40** in phased antenna array **54** may be dielectric resonator antenna such as the probe-fed dielectric resonator antenna **40** of FIGS. **6-8** (e.g., having two feed probes **100V** and **100H** as shown in FIG. **6** and optionally parasitic elements **108V** and **108H** as shown in FIG. **7** or having one feed probe **100** and optionally parasitic elements **114-1** and **114-2** as shown in FIG. **8**). Phased antenna array **54** may be formed as a part of an integrated antenna module.

(98) FIG. **9** is a top down view of an integrated antenna module that may include phased antenna array **54**. As shown in FIG. **9**, phased antenna array **54** may be formed as a part of an integrated antenna module such as antenna module **120**. Antenna module **120** may include substrate **72**. Phased antenna array **54** may be mounted to a surface of substrate **72** such as surface **122**. A board-to-board connector such as connector **123** may also be mounted to surface **122**.

(99) In the example of FIG. **9**, phased antenna array **54** is a dual-band phased antenna array having a first set of antennas **40L** that convey radio-frequency signals in a first frequency band and a second set of antennas **40H** that convey radio-frequency signals in a second frequency band that is higher than the first frequency band. Antennas **40H** may therefore sometimes be referred to herein as high band antennas **40H** whereas low band antennas **40L** are sometimes referred to herein as low band antennas **40L**. As just one example, the first frequency band may include frequencies between about 24 and 31 GHz and the second frequency band may include frequencies between about 37 and 41 GHz.

(100) High band antennas **40H** may be dielectric resonator antennas having dielectric resonating elements **68H** embedded within dielectric substrate **70**. Low band antennas **40L** may be dielectric resonator antennas having dielectric resonating elements **68L** embedded within dielectric substrate **70**. Dielectric substrate **70** may be molded over and/or around dielectric resonating elements **68H** and **68L** and may be mounted to surface **122** of substrate **72**. In order to support satisfactory beam forming, each high band antenna **40H** may, for example, be separated from one or two adjacent high band antennas **40H** in dielectric substrate **70** by a distance that is approximately equal to one-half of the effective wavelength corresponding to a frequency in the second frequency band (e.g., where the effective wavelength is equal to a free space wavelength multiplied by a constant value determined by the dielectric material surrounding the antennas). Similarly, each low band antenna **40L** may, for example, be separated from one or two adjacent low band antennas **40L** in dielectric substrate **70** by a distance that is approximately equal to one-half of the effective wavelength corresponding to a frequency in the first frequency band.

(101) In the example of FIG. **9**, phased antenna array **54** is a one-dimensional array having four high band antennas **40H** interleaved (interspersed) with four low band antennas **40L** arranged along a single longitudinal axis (e.g., running parallel to the X-axis). This is merely illustrative. Phased antenna array **54** may include any desired number of low band antennas **40L** and/or high band antennas **40H** and the antennas may be arranged in any desired one or two-dimensional pattern.

(102) FIG. **10** is a cross-sectional side view of antenna module **120** (e.g., as taken in the direction of line AA' of FIG. **9**). As shown in FIG. **10**, the bottom surface **82** of the dielectric resonating elements **68L** and **68H** in phased antenna array **54** may be mounted to surface **122** of substrate **72**. Dielectric substrate **70** may be molded over dielectric resonating elements **68L** and **68H** and may be mounted to surface **122**. If desired, dielectric substrate **70** may be molded over every dielectric resonating element **68L** and **68H** in phased antenna array **54** to form a single integrated structure, and the single integrated structure may then be mounted (e.g., surface-mounted) to surface **122** of substrate **72**. This may, for example, minimize mechanical variations between the antennas in phased antenna array **54** that could otherwise deteriorate antenna performance or mechanical reliability.

(103) Substrate **72** may have a surface **124** opposite surface **122**. Additional electronic components such as radio-frequency integrated circuit (RFIC) **126** may be mounted to surface **124** of substrate **72**. An optional over-mold and/or shielding structures may be provided over RFIC **126** and surface

124 of substrate **72** (not shown in the example of FIG. **10** for the sake of clarity). RFIC **126** may have terminals or ports that are coupled to corresponding contact pads on surface **124** using solder balls, conductive adhesive, conductive pins, conductive springs, and/or any other desired conductive interconnect structures.

(104) Radio-frequency transmission lines in substrate **72** (e.g., radio-frequency transmission lines **88V** and **88H** of FIG. **6**) may couple the ports of RFIC **126** to the feed probes (e.g., feed probes **100V** and **100H** of FIG. **6**) on dielectric resonating elements **68L** and **68H**. Dielectric substrate **72** may include multiple stacked dielectric substrate layers (e.g., layers of printed circuit board material, flexible printed circuit material, ceramic, etc.). The radio-frequency transmission lines in substrate **72** may include signal traces and ground traces on one or more of the stacked dielectric substrate layers (e.g., embedded within and/or on surfaces **122** and/or **124** of substrate **72**) and/or conductive vias extending through one or more of the stacked dielectric substrate layers.

(105) RFIC **126** may include, for example, phase and magnitude controllers **50** of FIG. **4**, upconverter circuitry, down-converter circuitry, amplifier circuitry, or any other desired radio-frequency circuitry. RFIC **126** may include one or more additional ports or terminals that are coupled to connector **123** of FIG. **9** (e.g., using additional radio-frequency transmission line structures on substrate **72**). RFIC **126** may be coupled to millimeter/centimeter wave transceiver circuitry **38** of FIGS. **2** and **3** via connector **123**. Millimeter/centimeter wave transceiver circuitry **38** may be mounted to an additional substrate such as an additional rigid printed circuit board, a flexible printed circuit, the main logic board of device **10**, etc. If desired, the signals conveyed between the millimeter/centimeter wave transceiver circuitry and RFIC **126** may be at an intermediate frequency (e.g., a radio frequency) that is greater than a baseband frequency and less than the frequencies with which antennas **40L** and **40H** convey radio-frequency signals. In these scenarios, upconverter circuitry in RFIC **126** may up-convert the signals from the intermediate frequency to the frequencies of operation of antennas **40L** and **40H**. Similarly, downconverter circuitry in RFIC **126** may down-convert signals received by antennas **40L** and **40H** to the intermediate frequency. RFIC **126** may, if desired, include multiple separate (discrete) radio-frequency integrated circuits mounted to substrate **72** (e.g., antenna module **120** may be an integrated circuit package that includes one or more RFICs and one or more phased antenna arrays mounted to a common/shared substrate such as substrate **72**).

(106) FIG. **11** is a perspective view of the antenna module **120** of FIGS. **9** and **10**. As shown in FIG. **11**, phased antenna array **54** (e.g., dielectric resonating elements **68L** and **68H** and dielectric substrate **70**) may be mounted to surface **122** of substrate **72**. Dielectric substrate **70** may have a foot structure **128** at surface **122** that is wider than the top surface of dielectric substrate **70** (e.g., to increase the mechanical stability of antenna module **120**). If desired, phased antenna array **54** may be secured to surface **122** using a layer of adhesive. Underfill may be provided under dielectric substrate **70** and phased antenna array **54** if desired. In the example of FIG. **11**, a dielectric over-mold structure such as over-mold **131** is provided on surface **124** of substrate **72**. Over-mold **131** may cover RFIC **126** of FIG. **10** (e.g., RFIC **126** may be embedded within over-mold **131**, thereby hiding RFIC **126** from view in FIG. **11**). Over-mold **131** may serve to protect RFIC **126** from damage or contaminants, may perform heat dissipation, isolation, shielding, etc. Phased antenna array **54** may be mounted within peripheral region **66** of FIG. **5** and may convey radio-frequency signals through the front or rear face of device **10**, as examples.

(107) In the example of FIGS. **9-11**, RFIC **126** is mounted to the opposite side of substrate **72** as phased antenna array **54**. This is merely illustrative. In another suitable arrangement, RFIC **126** may be mounted to the same side of substrate **72** as phased antenna array **54**. FIG. **12** is a top-down view showing how RFIC **126** may be mounted to the same side of substrate **72** as phased antenna array **54**.

(108) As shown in FIG. **12**, RFIC **126** and phased antenna array **54** may both be mounted to surface **122** of substrate **72**. Some or all of RFIC **126** may, for example, be laterally interposed

between phased antenna array 54 and a peripheral edge of substrate 72. FIG. 13 is a side view of antenna module 120 as taken in the direction of arrow 132 of FIG. 12. As shown in FIG. 13, phased antenna array 54 may be taller in the direction of the Z-axis than RFIC 126. This may, for example, allow RFIC 126 to rest under display module 64 while phased antenna array 54 radiates through display cover layer 56 (e.g., in scenarios where antenna module 120 is mounted within peripheral region 66 of FIG. 5 and phased antenna array 54 is a front-facing phased antenna array in device 10).

(109) If desired, antenna module 120 may include multiple phased antenna arrays mounted to different sides of substrate 72. FIG. 14 is a side view showing how multiple phased antenna arrays 54 may be mounted to different sides of substrate 72. As shown in FIG. 14, antenna module 120 may include a first phased antenna array 54-1 and a second phased antenna array 54-2. First phased antenna array 54-1 may include antennas 40 with dielectric resonating elements 68 mounted to surface 122 of substrate 72 whereas second phased antenna array 54-2 includes antennas 40 with dielectric resonating elements 68 mounted to surface 124 of substrate 72. First phased antenna array 54-1 may steer a beam of radio-frequency signals 134 across at least some of the hemisphere above surface 122. Second phased antenna array 54-2 may steer a beam of radio-frequency signals 136 across at least some of the hemisphere below surface 124. First phased antenna array 54-1 may be a one-dimensional array or a two-dimensional array of antennas 40. Second phased antenna array 54-2 may be a one-dimensional array or a two-dimensional array of antennas 40.

(110) Antenna module 120 of FIG. 14 may, for example, be mounted within peripheral region 66 of FIG. 5. First phased antenna array 54-1 may be a front-facing phased antenna array (e.g., where radio-frequency signals 134 serve as the radio-frequency signals 62 conveyed through display cover layer 56 of FIG. 5). Second phased antenna array may be a rear-facing phased antenna array (e.g., where radio-frequency signals 136 serve as the radio-frequency signals 60 conveyed through rear housing wall 12R of FIG. 5). In another suitable arrangement, first phased antenna array 54-1 may be a rear-facing phased antenna array whereas second phased antenna array 54-2 is a front-facing phased antenna array.

(111) As shown in FIG. 14, connector 123 may be mounted to surface 122. This is merely illustrative and, in another suitable arrangement, connector 123 may be mounted to surface 124. RFIC 126 may be mounted to surface 124. This is merely illustrative and, in another suitable arrangement, RFIC 126 may be mounted to surface 122. RFIC 126 and connector 123 may be mounted to the same surface if desired. Radio-frequency transmission lines in substrate 72 may couple RFIC 126 to each of the antennas 40 in phased antenna arrays 54-1 and 54-2. An over-mold structure may be provided over RFIC 126 and surface 124 if desired. In the example of FIG. 14, phased antenna arrays 54-1 and 54-2 are shown without a corresponding dielectric substrate 70 (FIGS. 6 and 9-13) for the sake of clarity. If desired, dielectric substrates 70 may be molded over first phased antenna array 54-1 and/or second phased antenna array 54-2.

(112) The example of FIG. 14 in which both phased antenna arrays 54-1 and 54-2 are formed from dielectric resonator antennas is merely illustrative. In another suitable arrangement, the antennas in first phased antenna array 54-1 may be stacked patch antennas, as shown in the cross-sectional side view of FIG. 15.

(113) As shown in FIG. 15, the antennas 40 in first phased antenna array 54-1 may be stacked patch antennas. Each antenna 40 in first phased antenna array 54-1 may include one or more conductive patches 140 embedded within the dielectric layers 138 of substrate 72. Conductive patches 140 may be spaced apart from and extend parallel to ground traces 144 in substrate 72. The conductive patches 140 in antennas 40 may include directly-fed patch antenna resonating elements and/or indirectly-fed parasitic antenna resonating elements that at least partially overlap at least one directly-fed patch antenna resonating element. Conductive patches 140 may have lengths 142 that determine the frequency response of first phased antenna array 54-1. Lengths 142 may, for example, be approximately equal to one-half the effective wavelength corresponding to a frequency

in the frequency band of operation of first phased antenna array **54-1**.

(114) In practice, the dielectric resonating elements **68** in second phased antenna array **54-2** may occupy greater height (e.g., in the direction of the Z-axis) than conductive patches **140** in first phased antenna array **54-1**. At the same time, conductive patches **140** may occupy greater area (e.g., in the X-Y plane) than dielectric resonating elements **68**. This may allow antenna module **120** to be mounted within device **10** at locations where there may be more space to place antennas for radiating through one side of device **10** than the other. As an example, antenna module **120** of FIG. **15** may be mounted within peripheral region **66** of FIG. **5** with second phased antenna array **54-2** facing display cover layer **56** and first phased antenna array **54-1** facing rear housing wall **12R** (e.g., there may be more space to place antennas for radiating through rear housing wall **12R** than through display cover layer **56** due to the presence of display module **64**). The example of FIG. **15** is merely illustrative and, in another suitable arrangement, first phased antenna array **54-1** may include dielectric resonator antennas whereas second phased antenna array **54-2** includes stacked patch antennas.

(115) In practice, it can be challenging to manufacture antenna modules having dielectric resonator antennas such as antenna module **120** of FIGS. **9-15**. In some scenarios, antenna modules are manufactured by individually forming each dielectric resonating element (e.g., by sintering a ceramic powder), individually metallizing the probe feed for each dielectric resonating element, injection molding the dielectric substrate over each individually-formed dielectric resonating element in the array, grinding down the portion of the dielectric resonating elements protruding beyond the dielectric substrate, and surface-mounting the result to a board. This process can be very complicated, time consuming, and expensive, and can lead to antenna modules that exhibit a substantial amount of mechanical variation that limits the overall mechanical and/or wireless performance of the module (e.g., due to poor dielectric resonating element parallelism, height coplanarity, and dimension, contact pad tolerance issues, and unpredictable dielectric resonating element tilting). In order to mitigate these issues, antenna module **120** may be manufactured using a largely scalable, IC-assembly process compatible, double side molding process, as shown in FIGS. **16** and **17**.

(116) FIGS. **16** and **17** are diagrams of an illustrative assembly process for antenna module **120**. As shown in FIG. **16**, antenna modules **120** may be manufactured in a manufacturing system such as manufacturing system **146**. Manufacturing system **146** may include manufacturing equipment **148**. Manufacturing system **146** may gather substrate **72** and electronic components **150** to be assembled into a given antenna module **120**. Substrate **72** may include radio-frequency transmission line structures (e.g., signal and ground traces on or embedded within the dielectric layers of substrate **72**) and corresponding contact pads coupled to the radio-frequency transmission line structures at the surfaces of substrate **72**. Electronic components **150** may include RFIC **126** (FIG. **9-15**) or any other desired radio-frequency components (e.g., radio-frequency switching circuits, filter circuits, discrete capacitors, resistors, and inductors, amplifier circuits, etc.).

(117) Manufacturing equipment **148** may surface mount electronic components **150** to surface **122** of substrate **72**, as shown by arrow **152** (e.g., using surface-mount technology (SMT) equipment in manufacturing equipment **148**). For example, solder balls **154** or any other desired conductive interconnect structures may be used to couple the terminals (ports) of electronic components **150** to corresponding contact pads on surface **122** of substrate **72**. Manufacturing equipment **148** may then layer over-mold **131** over the surface-mounted components **150** and surface **122** of substrate **72**, as shown by arrow **156**. This may serve to encapsulate or embed electronic components **150** at surface **122** within over-mold **131**.

(118) Manufacturing equipment **148** may then flip substrate **72** over and each dielectric resonating element **68** in the antenna module may be concurrently formed on surface **124** of substrate **72**. For example, manufacturing equipment **148** may form dielectric resonating elements **68** by performing a molding/selective molding process using high dielectric constant epoxy mold compound material

to mold each of the dielectric resonating elements **68** in the module at once (e.g., so that dielectric resonating elements **68** exhibit dielectric constant ϵ_k1 of FIG. **6**). This process may also form a top-most layer **164** on surface **124** of substrate **72**. Top-most layer **164** may cover the contact pads at surface **124** for the radio-frequency transmission lines used to feed dielectric resonating elements **68** (e.g., radio-frequency transmission lines **88V** and **88H** of FIG. **6**). While top-most layer **164** may be formed from the same material as dielectric resonating elements **68**, top-most layer **164** may sometimes be referred to herein as forming a part of substrate **72** or forming the top-most layer of substrate **72**.

(119) Manufacturing equipment **148** may then perform laser activation and metallization for dielectric resonating elements **68** (e.g., using a laser direct structuring (LDS) process), as shown by arrow **162**. For example, lasers in manufacturing equipment **148** may be used to create a pattern or seed layer for the metallization of the feed probes and optionally the parasitic elements for antennas **40** (e.g., on sidewalls **102** of dielectric resonating elements **68** and/or on top-layer **164**).

Manufacturing equipment **148** may then perform a physical deposition or chemical plating process that metalizes the pattern or seed layer created by the lasers. This may serve to form conductive structures **86V** and **86H** on sidewalls **102** of dielectric resonating elements **68** (e.g., at bottom surface **82** of dielectric resonating elements **68**) and/or on top-most layer **164**. If desired, this process may also be used to form parasitic elements **108H** and **108V** (FIG. **7**) and/or parasitic elements **114-1** and **114-2** (FIG. **8**) on sidewalls **102** and/or top-most layer **164**. In scenarios where dielectric resonating elements **68** only cover a single polarization, manufacturing equipment **148** may form only a single feed probe on each dielectric resonating element **68**.

(120) In addition, manufacturing equipment **148** may couple conductive structures **86V** and **86H** to corresponding contact pads on surface **124** of substrate **72** (e.g., by forming conductive vias that extend through top-most layer **164**). In scenarios where parasitic elements **114-1** and/or **114-2** of FIG. **8** are formed, manufacturing equipment **148** may form conductive vias through top-most layer **164** to couple the parasitic elements to ground traces at surface **124**. Coupling conductive structures **86V** and **86H** to the contact pads on surface **124** may serve to couple conductive structures **86V** and **86H** to corresponding radio-frequency transmission lines in substrate **72**. The radio-frequency transmission lines may couple conductive structures **86V** and **86H** to electronic components **150** at surface **122**.

(121) If desired, multiple antenna modules **120** may be manufactured from the same substrate **72**, as shown in the perspective view of FIG. **17**. As shown in FIG. **17**, substrate **72** may be used to form nine antenna modules each having four antennas and thus four dielectric resonating elements **68** arranged in a 1×4 pattern. This example is merely illustrative and, in general, any desired number of antenna modules may be formed from the same substrate **72**. The processes of FIG. **16** may be performed concurrently for each of the antenna modules formed from substrate **72**.

Concurrently manufacturing multiple antenna modules in this way may increase the reliability of the antenna modules (both within each antenna module and between antenna modules) and reduce the cost and time required to manufacture multiple devices **10**. This process may allow antenna module **120** to exhibit a smaller form factor for multiple applications, may eliminate extra injection molding, sintering, surface-mounting, and underfilling relative to arrangements where each dielectric resonating element is individually molded and then mounted to a substrate. This arrangement may also allow for tighter process control and improved yield relative to arrangements where each dielectric resonating element is individually molded and then mounted to a substrate.

(122) As by arrow **166**, manufacturing equipment **148** may surface-mount connectors **123** to connector contact pads **168** at surface **124** of substrate **72**. Connectors **123** may couple electronic components **150** in over-mold **131** to transceiver circuitry on a separate substrate after the antenna modules are assembled into device **10**, for example. Cutting equipment (e.g., blade or laser cutting tools) in manufacturing equipment **148** may then dice (cut) substrate **72** into separate antenna modules, as shown by arrow **170**. In the example of FIG. **17**, this may produce nine separate strips

of substrate **72** that form nine separate antenna modules **120**, each having four antennas **40** with corresponding dielectric resonating elements **68**. Dielectric structure **70** may be molded over dielectric resonating elements **68** after dicing, at any other desired time after conductive structures **86H** and **86V** have been formed on dielectric resonating elements **68**, or may be omitted if desired. (123) FIG. **18** is a flow chart of illustrative steps that may be performed by manufacturing equipment **148** of FIGS. **16** and **17** in manufacturing antenna module **120**. At step **172**, manufacturing equipment **148** may surface-mount electronic components **150** (e.g., one or more radio-frequency integrated circuits) to a surface of substrate **72** (e.g., as shown by arrow **152** of FIG. **16**). Manufacturing equipment **148** may layer over-mold **131** over the surface-mounted electronic components **150** (e.g., as shown by arrow **156** of FIG. **16**).

(124) At step **174**, manufacturing equipment **148** may mold dielectric resonating elements **68** on a surface of substrate **72** (e.g., as shown by arrow **160** of FIG. **16**). Dielectric resonating elements **68** may be molded onto the surface of substrate **72** opposite to the surface-mounted electronic components **150**. This is merely illustrative and, if desired, dielectric resonating elements **68** may be molded onto the same surface of substrate **72** as the surface-mounted electronic components **150** (e.g., as shown in FIGS. **12** and **13**).

(125) At step **176**, manufacturing equipment **148** pattern conductive traces onto dielectric resonating elements **68** (e.g., as shown by arrow **162** of FIG. **16**). Manufacturing equipment **148** may, for example, use lasers to activate or create a seed layer on dielectric resonating elements **68**. Manufacturing equipment **148** may then deposit conductive material over the activated portions of dielectric resonating elements **68**. The conductive material may form conductive structures **86V** and **86H** (e.g., for feed probes **100V** and **100H** of FIG. **6**) and/or parasitic elements for the antennas.

(126) At step **178**, manufacturing equipment **148** may surface-mount connectors **123** onto the connector contact pads **168** of substrate **72** (e.g., as shown by arrow **166** of FIG. **17**).

(127) At step **180**, manufacturing equipment **148** may dice substrate **180** into individual antenna modules **120** and may add corresponding shielding structures to the antenna modules (e.g., as shown by arrow **170** of FIG. **17**). The shielding may serve to isolate electronic components **150** from electromagnetic interference, for example.

(128) At step **182**, manufacturing equipment **148** may assemble a manufactured antenna module **120** into device **10**. For example, manufacturing equipment **148** may mount antenna module **120** within peripheral region **66** of FIG. **5** or elsewhere within the interior of device **10**. Antenna module **120** may be mounted to convey radio-frequency signals through display cover layer **56** or rear housing wall **12R** of FIG. **5**, for example. The steps of FIG. **18** are merely illustrative and, if desired, other processes may be used to manufacture antenna module **120**.

(129) In practice, implementation of dielectric resonator antennas in electronic devices can be challenging since the dielectric resonator antennas have high aspect ratios that make it difficult to control system alignment, reliability, and interconnect reliability. In other phased antenna arrays, each antenna may require two radio-frequency connectors to feed, which can be undesirably bulky. Integrating the dielectric resonator antennas into antenna module **120** may allow the antennas to each be fed without requiring as many connectors and may allow the antennas to be properly aligned with a high degree of reliability.

(130) In practice, the metallization used to feed dielectric resonating elements **68** can be costly to perform at scale. In another suitable arrangement, the feed probes for dielectric resonating elements **68** may be pressed against dielectric resonating elements **68** using feed probe biasing structures. This may allow the antennas to be fed without additional metallizations on the ceramic, which may decrease cost and design complexity.

(131) FIG. **19** is a perspective view of an illustrative antenna module **120** having feed probes that are pressed against dielectric resonating elements **68** using feed probe biasing structures. In the example of FIG. **19**, substrate **72** is a flexible printed circuit. Phased antenna array **54** may include

dielectric resonating elements **68** embedded within dielectric substrate **70** to form antenna package **184**. Antenna package **184** may then be surface-mounted to contact pads on surface **122** of substrate **72**. In the example of FIG. **19**, phased antenna array **54** includes two low band antennas **40L** interleaved with two high band antennas **40H** (e.g., in a 1×4 array). This is merely illustrative and, in general, phased antenna array **54** may include any desired number of antennas for covering any desired frequency bands. The antennas may be arranged in any desired pattern.

(132) As shown in FIG. **19**, the dielectric resonating element **68H** in high band antennas **40H** may be separated from the dielectric resonating element **68L** in one or two adjacent low band antennas **40L** by distance **192**. Distance **192** may be selected to provide satisfactory electromagnetic isolation between low band antennas **40L** and high band antennas **40H**. Each dielectric resonating element in phased antenna array **54** may be fed by feed probes having conductive structures **86V** and **86H**. Conductive structures **86V** and **86H** may be pressed against dielectric resonating elements **68** by feed probe biasing structures in antenna package **184** (not shown in FIG. **19** for the sake of clarity). The feed probe biasing structures may, for example, press or bias conductive structure **86H** against the sidewalls **102** of dielectric resonating elements **68** (e.g., by exerting a biasing force in the $-X$ direction). Similarly, the feed probe biasing structures may press or bias conductive structure **86V** against the sidewalls **102** of dielectric resonating elements **68** (e.g., by exerting a biasing force in the $+Y$ direction).

(133) Dielectric substrate **70** may be molded over the feed probe biasing structures as well as dielectric resonating elements **68**. Dielectric substrate **70** may have a bottom surface **188** at substrate **72** and an opposing top surface **190**. In the example of FIG. **19**, the top surface **80** of dielectric resonating elements **68** protrudes above top surface **190** of dielectric substrate **70**. This is merely illustrative and, if desired, top surface **190** may lie flush with the top surface **80**. In another suitable arrangement, dielectric substrate **70** may cover the top surface **80** of dielectric resonating elements **70**. An attachment structure **186** may be partially embedded within dielectric substrate **70** (e.g., dielectric substrate **70** may be molded over part of attachment structure **186**). Attachment structure **186** may help to secure antenna module **120** in place within device **10** if desired (e.g., using screws, pins, or other structures that extend through an opening in attachment structure **186**).

(134) FIG. **20** is diagram of an illustrative assembly process for antenna module **120** of FIG. **19**. As shown in FIG. **20**, the antenna modules may be manufactured in manufacturing system **146**. Manufacturing equipment **148** may include alignment posts **194**. Alignment posts **194** may press conductive structure **86H** against a first sidewall **102** of dielectric resonating element **68** and may press conductive structure **86V** against a second (orthogonal) sidewall **102** of dielectric resonating element **68**. Conductive structures **86H** and **86V** may include stub portions **196** that lie in the X - Y plane. Conductive structures **86H** and **86V** may, for example, be stamped from pieces of sheet metal (e.g., while alignment posts press against conductive structures **86H** and **86V**, leaving behind stub portions **196**).

(135) This may allow for a tight control of the size and position of the stamped conductive structures **86H** and **86L** while minimizing gaps between the conductive structures and dielectric resonating element **68**.

(136) During a first molding process (e.g., a first injection molding process), manufacturing equipment **148** may mold a feed probe biasing structure such as biasing structure **200** (sometimes referred to herein as retention structure **200**) over sidewalls **102** and conductive structures **86H** and **86V** at bottom surface **82** of dielectric resonating element **68** (e.g., as shown by arrow **198**). Alignment posts **194** may hold conductive structures **86H** and **86V** in place during the first molding process and may be removed once biasing structure **200** has been formed (e.g., leaving behind alignment post holes **202** in biasing structure **200**). Once the alignment posts **194** have been removed, biasing structure **200** may hold conductive structures **86V** and **86H** in place against the sidewalls **102** of dielectric resonating element **68**. Biasing structure **200** may, for example, exert a biasing force in the $-X$ direction against conductive structure **86H** and may exert a biasing force in

the +Y direction against conductive structure **86V**. Stub portions **196** of conductive structures **86H** and **86V** may remain exposed after molding biasing structure **200** onto dielectric resonating element **68**. This may allow stub portions **196** to be coupled to corresponding contact pads at surface **122** of substrate **72** of FIG. **19** (e.g., using solder, conductive adhesive, etc.), thereby forming the feed probes for antenna module **120**. Biasing structure **200** may have a bottom surface **206**. Bottom surface **206** may lie flush with bottom surface **82** of dielectric resonating element **68**. (137) This process may be performed for each antenna in antenna module **120**. Dielectric substrate **70** may subsequently be molded over each of the dielectric resonating elements **68**, the corresponding biasing structures **200**, and attachment structure **186** (e.g., using a second injection molding process) to form antenna package **184**, as shown by arrow **204**. For example, a tool in manufacturing equipment **148** may locate the over-molded dielectric substrate **70** by the plastic in biasing structures **200** to maintain the contact positions of conductive structures **86H** and **86V**. Dielectric substrate **70** may include one or more openings **208** (e.g., at locations where the tool in manufacturing equipment **148** held the dielectric resonating elements during over-molding). A spring feature on the tool may locate the top surface **80** of dielectric resonating elements **68** to prevent shifting during molding, thereby maintaining reliable coplanarity for the bottom surface **82** across each dielectric resonating element **68** in antenna package **184** (e.g., bottom surface **206** of biasing structures **200** may be coplanar with bottom surface **82** of dielectric resonating elements **68L** and **68H**, stub portions **196** of conductive structures **86H** and **86V**, and bottom surface **188** of dielectric substrate **70** across antenna package **184** with a very tight tolerance). This uniform and reliable coplanarity may allow the bottom surface of antenna package **184** to be surface-mounted to substrate **72** (thereby forming antenna module **120**) with minimal or uniform gaps across antenna package **184**, thereby optimizing the mechanical reliability and wireless performance of antenna module **120**. Antenna module **120** may then be mounted within device **10**.

(138) FIG. **21** is a top-down view showing one illustrative location where antenna module **120** may be mounted within device **10** (e.g., antenna module **120** of FIG. **19** or other antenna modules **120** as described herein). As shown in FIG. **21**, display module **64** in display **14** may include notch **8**. Display cover layer **56** of FIG. **5** has been omitted from FIG. **21** for the sake of clarity. Display module **64** may form active area AA of display **14** whereas notch **8** forms part of inactive area IA of display **14** (FIG. **1**). The edges of notch **8** may be defined by peripheral conductive housing structures **12W** and display module **64**. For example, notch **8** may have two or more edges (e.g., three edges) defined by display module **64** and one or more edges defined by peripheral conductive housing structures **12W**.

(139) Device **10** may include speaker port **16** (e.g., an ear speaker) within notch **8**. If desired, device **10** may include other components **210** within notch **8**. Other components **210** may include one or more image sensors such as one or more cameras, an infrared image sensor, an infrared light emitter (e.g., an infrared dot projector and/or flood illuminator), an ambient light sensor, a fingerprint sensor, a capacitive proximity sensor, a thermal sensor, a moisture sensor, or any other desired input/output components (e.g., input/output devices **26** of FIG. **2**). Antenna module **120** (e.g., an antenna module having dielectric resonating elements **68L** interleaved with dielectric resonating elements **68H** for covering different frequency bands) may be mounted within device **10** (e.g., within peripheral region **66** of FIG. **5**) and aligned with the portion(s) of notch **8** that are not occupied by other components **210** or speaker port **16**. Antenna module **120** may be laterally interposed between two components **210** such as between an image sensor (e.g., a rear-facing camera) and an ambient light sensor, dot projector, flood illuminator, or ambient light sensor, for example.

(140) Substrate **72** may extend under display module **64** to another substrate such as substrate **214** (e.g., another flexible printed circuit, a rigid printed circuit board, a main logic board, etc.). The radio-frequency transceiver circuitry for antenna module **120** may be mounted to substrate **214** if desired. Connector **123** on substrate **72** may be coupled to connector **212** (e.g., a board-to-board

connector) on substrate **214**. This may allow the antennas in antenna module **120** to cover at least some of the hemisphere over the front face of device **10** without occupying an excessive amount of space within device **10**, for example. The example of FIG. **21** is merely illustrative and, in general, antenna module **120** may be mounted at any desired location within device **10**. Antenna module **120** may have any desired number of antennas for covering any desired frequency bands. The antennas in antenna module **120** may be arranged in any desired one or two-dimensional pattern. (141) In order to further increase isolation between adjacent antennas **40** in phased antenna array **54**, each dielectric antenna resonating element in the array may be rotated relative to as shown in FIGS. **9-21**. FIG. **22** is a top view showing how phased antenna array **54** may include rotated dielectric antenna resonating elements.

(142) As shown in FIG. **22**, antenna module **120** may include dielectric resonating elements **68H** and **68L** that are arranged in a one-dimensional pattern along longitudinal axis **216** (e.g., an axis running through the central/longitudinal axis of each of the dielectric resonating elements). Dielectric substrate **70** may be molded over dielectric resonating elements **68H** and **68L**. Prior to mounting to substrate **72**, dielectric resonating elements **68H** and **68L** may be rotated so that the sidewalls of the dielectric resonating elements (e.g., the lateral/peripheral edges of the dielectric resonating elements as viewed from above) are each oriented at a non-parallel angle with respect to longitudinal axis **216**. For example, each dielectric resonating element **68H** and **68L** may include a first pair of opposing sidewalls **102** that are oriented at angle θ with respect to longitudinal axis **216**. Each dielectric resonating element **68H** and **68L** may also include a second pair of opposing sidewall **102** that are oriented perpendicular to the first pair of opposing sidewall (e.g., at a 90 degree angle with respect to the first pair of opposing sidewalls or an angle of angle $\theta+90$ degrees with respect to longitudinal axis **216**). In this way, the sidewalls may also be oriented at a non-parallel angle with respect to each lateral edge of substrate **72**, if desired. Angle θ may be between 0 degrees and 90 degrees (e.g., 45 degrees, 30-60 degrees, 40-50 degrees, etc.). Orienting dielectric resonating elements **68L** and **68H** in this way may serve to minimize cross-coupling between adjacent antennas **40L** and **40H**, thereby maximizing isolation between the antennas and thus the radio-frequency performance of antenna module **120**.

(143) In the example of FIG. **22**, phased antenna array **54** includes four low band antennas **40L** interleaved with four high band antennas **40H**. This example is merely illustrative. In general, phased antenna array **54** may include any desired number of antennas for covering any desired bands and arranged in any desired one or two-dimensional pattern on surface **122** of substrate **72**. Connector **123** may be mounted to surface **122** or the opposing surface of substrate **72**.

(144) FIG. **23** is a perspective view of the antenna module **120** of FIG. **22**. In the example of FIGS. **22** and **23**, the RFIC for antenna module **120** is mounted to surface **124** of substrate **72** and over-mold **131** is layered under surface **124** and the RFIC. This is merely illustrative and, in another suitable arrangement, the RFIC may be mounted to surface **122** (e.g., as shown in FIGS. **12** and **13**).

(145) As shown in FIG. **23**, feed probe biasing structures such as biasing structures **218** may press the feed probes for phased antenna array **54** against dielectric resonating elements **68L** and **68H** (e.g., by exerting biasing forces against the conductive structures in the feed probes that are oriented normal to the sidewalls **102** against which the feed probes are pressed). Dielectric substrate **70** may be molded over dielectric resonating elements **68L** and **68H** and biasing structures **218** (e.g., to form a single integrated antenna package that is then surface-mounted to substrate **72**). Dielectric substrate **70** may, if desired, include openings that expose biasing structures **218**. Dielectric substrate **70** may also include openings (holes) **219** that are laterally interposed between each pair of adjacent dielectric resonating elements in phased antenna array **54**. Openings **219** may, for example, serve to further increase isolation between the antennas **40L** and **40H** in phased antenna array **54**.

(146) FIG. **24** is an exploded view of the antenna module **120** of FIGS. **22** and **23**. As shown in

FIG. 24, feed probes **100V** and **100H** and optionally parasitic elements **108** may be pressed against dielectric resonating elements **68L** and **68H** by biasing structures **218**. In scenarios where dielectric resonating elements **68L** and **68H** are fed by only a single feed probe, parasitic elements **108** may be omitted and/or parasitic elements **114-1** and **114-2** of FIG. 8 may be used.

(147) Biasing structure **218** may be molded over dielectric resonating elements **68L** and **68H**, feed probes **100H** and **100V**, and parasitic elements **108** during a first molding process (e.g., similar to the first molding process associated with arrow **198** of FIG. 20). Alignment posts may press feed probes **100H** and **100V** and parasitic elements **108** against the dielectric resonating elements during the first molding process and may leave behind alignment post openings in biasing structures **218** after molding. Biasing structures **218** may press feed probes **100H** and **100V** and parasitic elements **108** against dielectric resonating elements **68L** and **68H** to maintain a reliable coupling between the feed probes, parasitic elements, and the dielectric resonating elements. Dielectric substrate **70** may be molded over all of the dielectric resonating elements **68H** and **68L** and biasing structures **218** in a second molding process (e.g., similar to the second molding process associated with arrow **204** of FIG. 20). The assembled phased antenna array **54** may subsequently be surface-mounted to substrate **72** of FIGS. 22 and 23 to form antenna module **120**.

(148) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a printed circuit board; a phased antenna array having dielectric resonating elements mounted to a surface of the printed circuit board; feed probes coupled to the dielectric resonating elements and configured to excite the dielectric resonating elements; and a plastic substrate on the printed circuit board, wherein the dielectric resonating elements are embedded in the plastic substrate, the dielectric resonating elements comprise first dielectric resonating elements having a first width and second dielectric resonating elements having a second width different than the first width, and the second dielectric resonating elements are interleaved with the first dielectric resonating elements.
2. The electronic device of claim 1, wherein the plastic substrate comprises injection-molded plastic.
3. The electronic device of claim 2, wherein the dielectric resonating elements comprise ceramic columns.
4. The electronic device of claim 1, wherein the plastic substrate has a first surface at the printed circuit board and a second surface opposite the first surface, the dielectric resonating elements extending from the first surface to the second surface of the plastic substrate.
5. The electronic device of claim 1, wherein the printed circuit board comprises a flexible printed circuit having transmission lines coupled to the feed probes.
6. The electronic device of claim 1, wherein the dielectric resonating elements are arranged on the printed circuit board along a longitudinal axis.
7. The electronic device of claim 6, wherein the first dielectric resonating elements and the second dielectric resonating elements have sidewalls oriented at a non-parallel angle and non-perpendicular angle with respect to the longitudinal axis.
8. The electronic device of claim 1, further comprising openings in the plastic substrate and laterally interposed between the dielectric resonating elements.
9. The electronic device of claim 1, further comprising: a display having an active area, an inactive area, and a display cover layer overlapping the active area and the inactive area, the inactive area including a notch in the active area; and peripheral conductive housing structures extending around a periphery of the display, wherein the display is mounted to the peripheral conductive housing

structures, the dielectric resonating elements overlap the notch, and the phased antenna array is configured to convey radio-frequency signals through the display cover layer.

10. An electronic device comprising: a printed circuit board; a phased antenna array having dielectric columns mounted to a surface of the printed circuit board, wherein the dielectric columns have first surfaces at the printed circuit board, second surfaces opposite the first surfaces, and sidewalls extending from the first surfaces to the second surfaces, the dielectric columns are aligned along a longitudinal axis, and the sidewalls extend non-parallel and non-perpendicular with respect to the longitudinal axis; feed probes coupled to the dielectric columns and configured to excite a resonant mode of the dielectric columns; and a dielectric substrate on the printed circuit board and surrounding the dielectric columns, wherein the dielectric substrate has an opening that is interposed between and configured to isolate two adjacent dielectric columns of the dielectric columns.

11. The electronic device of claim 10, wherein the feed probes comprise metal on the sidewalls of the dielectric columns.

12. The electronic device of claim 11, wherein the metal is soldered to the printed circuit board.

13. The electronic device of claim 11, wherein the metal comprises folded sheet metal.

14. The electronic device of claim 13, wherein the folded sheet metal has a first portion pressed against the sidewalls and a second portion extending from the first portion and along a lateral surface of the printed circuit board.

15. The electronic device of claim 11, wherein the metal comprises conductive traces.

16. The electronic device of claim 10, further comprising: a display having an active area, an inactive area, and a display cover layer overlapping the active area and the inactive area, the inactive area including a notch in the active area; and peripheral conductive housing structures extending around a periphery of the display, wherein the display is mounted to the peripheral conductive housing structures, the dielectric columns overlap the notch, and the phased antenna array is configured to convey radio-frequency signals through the display cover layer.

17. An electronic device comprising: a printed circuit board; a phased antenna array having dielectric columns mounted to a surface of the printed circuit board, wherein the dielectric columns have first surfaces at the printed circuit board, second surfaces opposite the first surfaces, and sidewalls extending from the first surfaces to the second surfaces; antenna feeds coupled to the dielectric columns and configured to excite a resonant mode of the dielectric columns; and an injection-molded over-mold on the printed circuit board and laterally surrounding the dielectric columns, wherein the second surfaces of the dielectric columns protrude beyond the injection-molded over-mold.

18. The electronic device of claim 17, further comprising: a display having an active area, an inactive area, and a display cover layer overlapping the active area and the inactive area, the inactive area including a notch in the active area; and peripheral conductive housing structures extending around a periphery of the display, wherein the display is mounted to the peripheral conductive housing structures, the dielectric columns overlap the notch, and the phased antenna array is configured to convey radio-frequency signals through the display cover layer.

19. The electronic device of claim 17, wherein the dielectric columns comprise first dielectric columns configured to form a first antenna and second dielectric columns configured to form a second antenna.

20. The electronic device of claim 1, wherein the first dielectric resonating elements are configured to convey first radio-frequency signals in a first frequency band and the second dielectric resonating elements are configured to convey second radio-frequency signals in a second frequency band different than the first frequency band.
